

Authors
Queenie Fan Chang
Josefina Lo Prete

Supervisors
Ricardo Mayor Luque
Hamid Peiro

Thesis Cluster
Advanced
Materials

casting the in-between



casting the in-between

dynamic casting of functionally graded
geopolymers using urban excavation waste

Authors

Queenie Fan Chang
Josefina Lo Prete

Supervisors

Ricardo Mayor Luque
Hamid Peiro

External Advisors

Marcos Cruz
Areti Markopoulou

Institute for Advanced Architecture of Catalonia
Master in Advanced Architecture
Barcelona, Spain, September 2026



acknowledgements

This thesis would not have been possible without a few key people whose guidance and support shaped both the research and the journey behind it.

Ricardo and Hamid, for believing in the potential of this project.
Marcos, for constantly offering feedback and ideas.
Gabriel and Dani, for being friendly faces around the labs.
Isa, for being the ultimate cinematographer.
Cris and Ale, for cheering us on.



statement on the use of AI

Generative artificial intelligence tools were used during the preparation of this thesis solely as editorial and post-production assistance. For written content, these tools assisted with the refinement of grammar, clarity, and phrasing. AI-assisted image editing was also used in selected photographs for minor visual adjustments, such as background clean-up and presentation refinement. These modifications did not alter the content or outcomes represented in the original images.

All research, ideas, analysis, design work, experimental outcomes, and original written and visual content presented in this thesis were developed and produced by the authors. The authors retained full responsibility for the development, interpretation, and final presentation of the work.



table of contents

10 abstract

12 chapter one

material and environmental framework

- 13 impacts of conventional construction
- 14 urban excavation material
 - 14 excavation material as urban resource
 - 15 digital fabrication constraints
- 15 alkali-activated materials

20 chapter two

material research

- 21 geopolymerizing urban excavation waste
 - 21 establishment of UEM as reaction precursor
- 26 compressive strength
- 29 material density
- 35 geometrical logic
 - 35 hyperbolic paraboloid
- 38 functional gradient
- 42 material research limitations
- 43 research question

44 chapter three

dynamic casting system development

- 45 hypothesis
- 46 state of the art
 - 47 dynamic casting
- 49 system design
 - 51 system parts
 - 52 evaluated parameters
 - 54 testing outcomes
- 62 prototype testing
 - 66 learning outcomes
 - 72 finalized system

78 chapter four

design application

- 79 assembly system
 - 81 component installation
- 81 case study: the cultural centre pavilion
 - 81 introduction
 - 82 architectural drawings
 - 86 assembly logic
 - 88 render images

94 chapter five

conclusion and future steps

100 bibliography



abstract

It is estimated that in the next 15 years, the world will generate over 3 billion tonnes of waste; in the European Union alone, 35% originates from the construction sector due to demolition, excavation, and dredging (“28 Incredible Statistics About Waste Generation,” 2021). These figures highlight the urgent need to rethink material systems in contemporary cities. Despite its abundance in urban contexts, excavation material is often discarded due to its heterogeneity, while current digital construction systems depend on homogeneous inputs. This thesis investigates how the variability of excavation material can be leveraged through functionally graded geopolymer systems. It asks: how can robotic fabrication enable the production of functionally graded, urban waste-based geopolymers? The research hypothesizes that robotic dynamic casting can transform heterogeneous, alkali-activated excavation material into freeform modular components with controlled material distribution and performance.

Positioned at the intersection of material reuse and fabrication innovation, the work operates on two fronts. First, it revalorizes excavation material as a primary construction resource through alkali activation. Second, it introduces a robotic dynamic casting system that enables material grading without extrusion-based constraints. Here, a flexible membrane formwork acts as an active agent, continuously reconfigured to shape geometry during casting. The hypothesis is tested through prototyping, material testing, and fabrication experiments using this dynamic casting method. Within this system, alkali-activated excavation material is cast into a flexible formwork in stages, enabling the production of freeform, functionally graded modular elements. The research ultimately proposes a scalable framework for integrating urban waste streams into high-performance architectural components, enabling a new architectural language shaped by material variability, robotic fabrication, and geometric freedom.



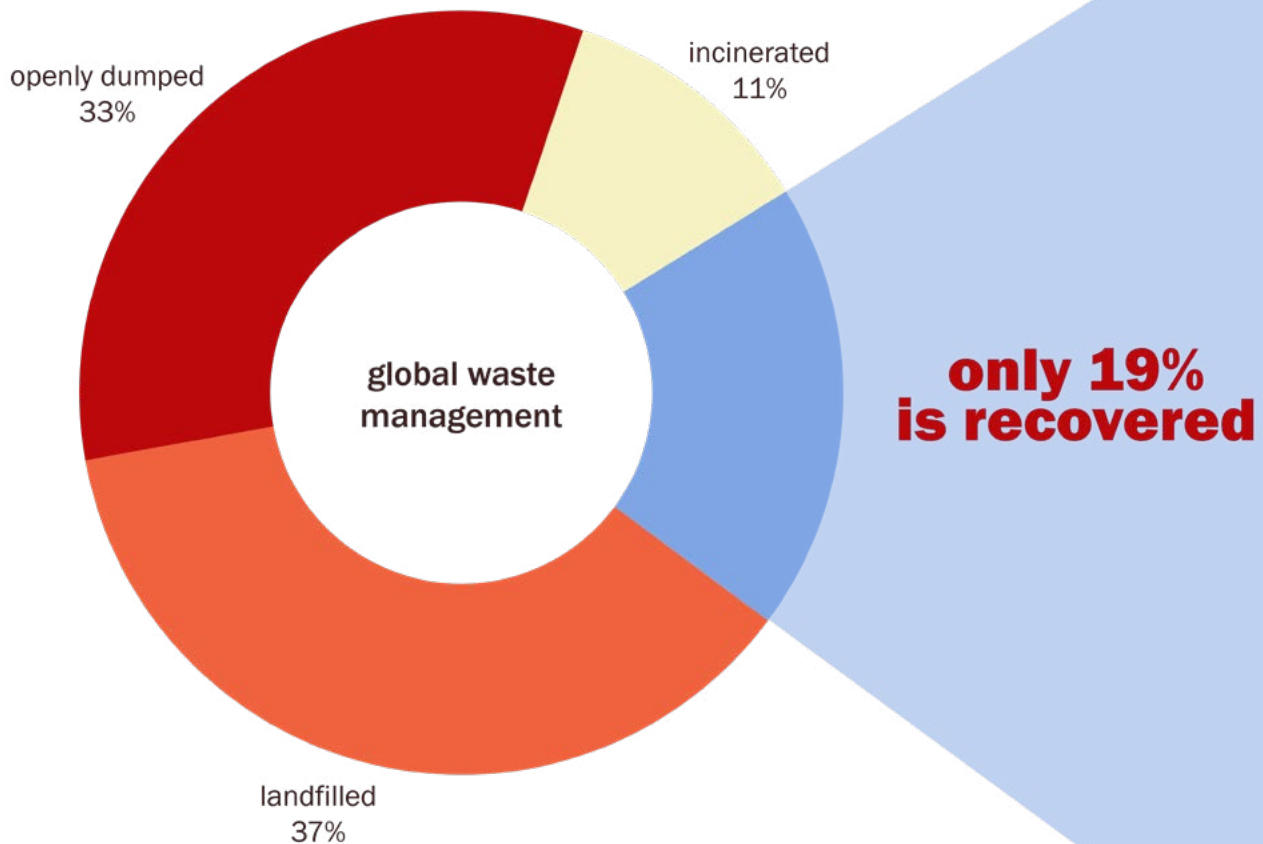
01

material and environmental framework

The motivation behind this research stems from the increasing generation of waste and the impact of the construction sector on the environment. Studies show that at the current rate, global annual waste generation is projected to reach approximately 3.4 billion tonnes by 2050 (FRONT Materials, 2021). Moreover, with current waste management, only about 19% of the waste generated is being recovered, while the remaining waste is either landfilled, dumped or incinerated (FRONT Materials, 2021).

Globally, there is a common pattern of behaviour and management practices in the construction industry, making it the largest consumer of natural resources. In the European Union alone, building construction consumes about 40% of material resources and generates 40% of all waste annually (Sizirici et al., 2021).(see Figure 1.1)

Figure 1.1 global waste management | Source: (FRONT Materials, 2021)



Impacts of Conventional Construction

As stated previously, from all this waste nearly 80% of it is left unused. As the population increases, these numbers will also increase. The global urban population (not to be confused with the overall global population) is projected to exceed 6 billion people in the next 20 years (Sizirici et al., 2021), meaning that without intervening as the demand for construction rises, so will all the material usage and waste generated.

Statistics show that within the waste generated in the Euro-

pean Union, nearly one third of it consists of construction and demolition waste (CDW) (FRONT Materials, 2021). This constitutes “all the waste produced by the construction and demolition of buildings and infrastructure, as well as road planning and maintenance, such as concrete, bricks, wood, glass, metals and plastic” (European Commission, n.d.). As one of the largest global waste streams that continues to grow as urban areas grow, there are many environmental implications that should be addressed. Not only is a lot of landfill capacity being unnecessarily

taken up, but there is also poor management of this waste that is not accounted for in databases meaning that the reported environmental impacts may be more underestimated than presented (Sönmez and Kalfa, 2023). One of the key ideas that is being presented is that CDW is not all “waste” material but rather contains large amounts of valuable resources that can be repurposed. Within this waste, about 80% of it consists of excavated earth, whether it is from construction sites, road construction or any other urban excavation (Sizirici et al., 2021).

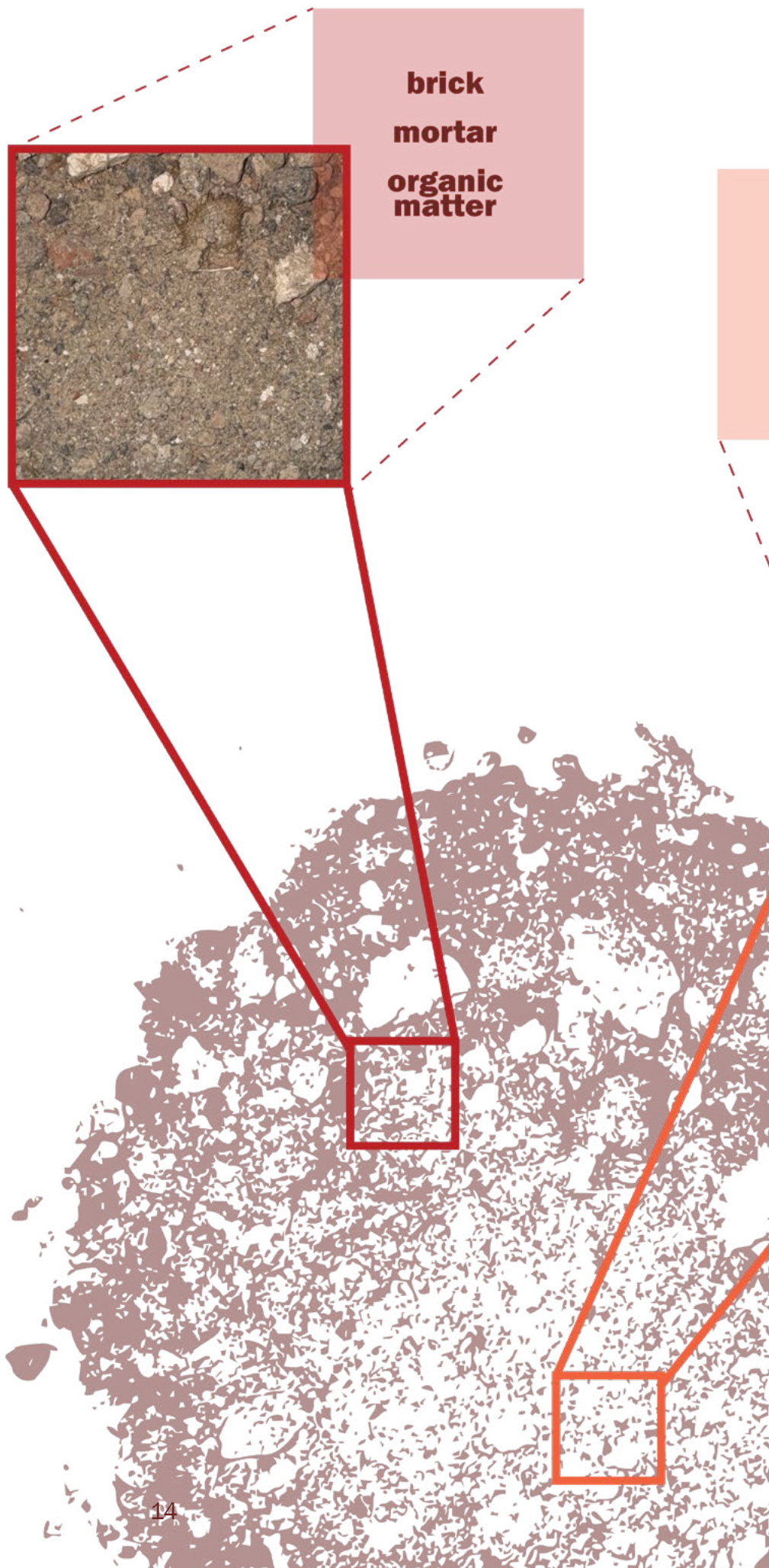
Urban Excavation Material

Urban excavation material (which moving forward will be referred to as UEM in this paper) is a mix of in-situ soil and rock removed from foundations, basements, tunnels, trenches, and other earthworks. It can also include gravel, sand, silt, clay, cobbles, weathered rock, and sometimes small amounts of organic matter and groundwater. The exact composition of the material is highly site-specific (IPA, 2019).

Excavation Material as Urban Resource

This material is typically land-filled or underused because it is classified as waste, may contain contaminants and is compositionally variable, making it difficult to process through conventional digital fabrication systems. Nonetheless, this thesis proposes to challenge this idea. There is material opportunity and potential in UEM. It is material that has already been extracted, so it avoids additional virgin-material extraction and diverts material from disposal; whether it is at a new construction site or during road repairs, for instance, this material was already going to be excavated. As locally extracted material, there is a significantly shorter supply chain that can lower transportation and sourcing costs in the fabrication process. Additionally, the excavated material's variability in composition gives it mineral diversity, which can provide different chemical and behavioural properties.

Figure 1.2 Urban excavation material typical composition, diagram



**asphalt
concrete
slag**



Digital Fabrication Constraints

Current digital fabrication systems rely on highly controlled and homogeneous material, particularly in additive manufacturing strategies. These are typically associated with extrusion-based and material deposition systems. A key property that makes a material extrudable, when it comes to digital fabrication, is the aggregate size. Printable mixes rely heavily on fine aggregates that all have similar particle size, which allows the material to properly flow through the piping and nozzle. When there is high variability in particle size, especially larger aggregates, there is a higher risk of clogging in the extrusion pipes (Rehman and Kim, 2021).

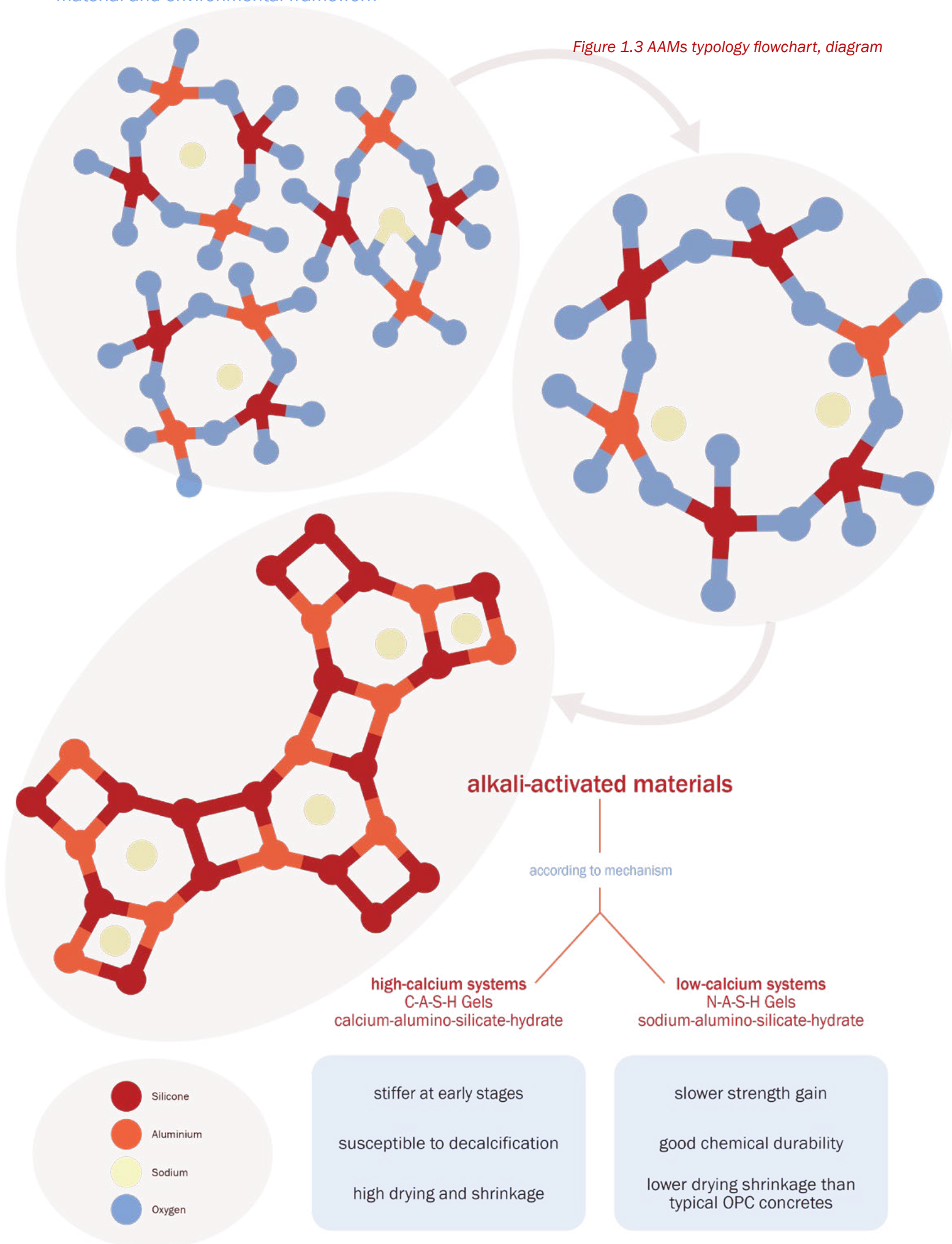
One of the key characteristics of UEM is the variability in its composition, and thus a high variability in its aggregate particle size, wherein there can be fine powder-like particles to large pieces of rock or broken concrete. As it stands, UEM does not fit the criteria to be extruded in current fabrication systems, not only due to its aggregate limitations but also because of its instability and unpredictability. So the ques-

tion is: **how can heterogeneous material be stabilised?**

Alkali-Activated Materials

Following the understanding of the composition and general behaviour of urban excavation material (UEM), the research shifted to finding a strategy to turn UEM into a buildable material, suitable for digital fabrication and capable of withstanding environmental exposure. Recent studies have shown that alkali-activated materials (AAMs) are being experimented with as an architectural material due to their similarities to traditional Portland cement (Qin et al., 2022). AAMs are cementitious binders produced from aluminosilicate precursors (typically industrial by-products) activated by alkaline solutions (Qin et al., 2022; Designing Buildings, 2023). As a binder, AAMs can then be combined with other aggregate sources that can bulk up the mix. This chemical reaction is a form of polymerization, where small molecules are chemically combined to create large chain-like networks of molecules, called polymers (Britannica Editors, 2026).

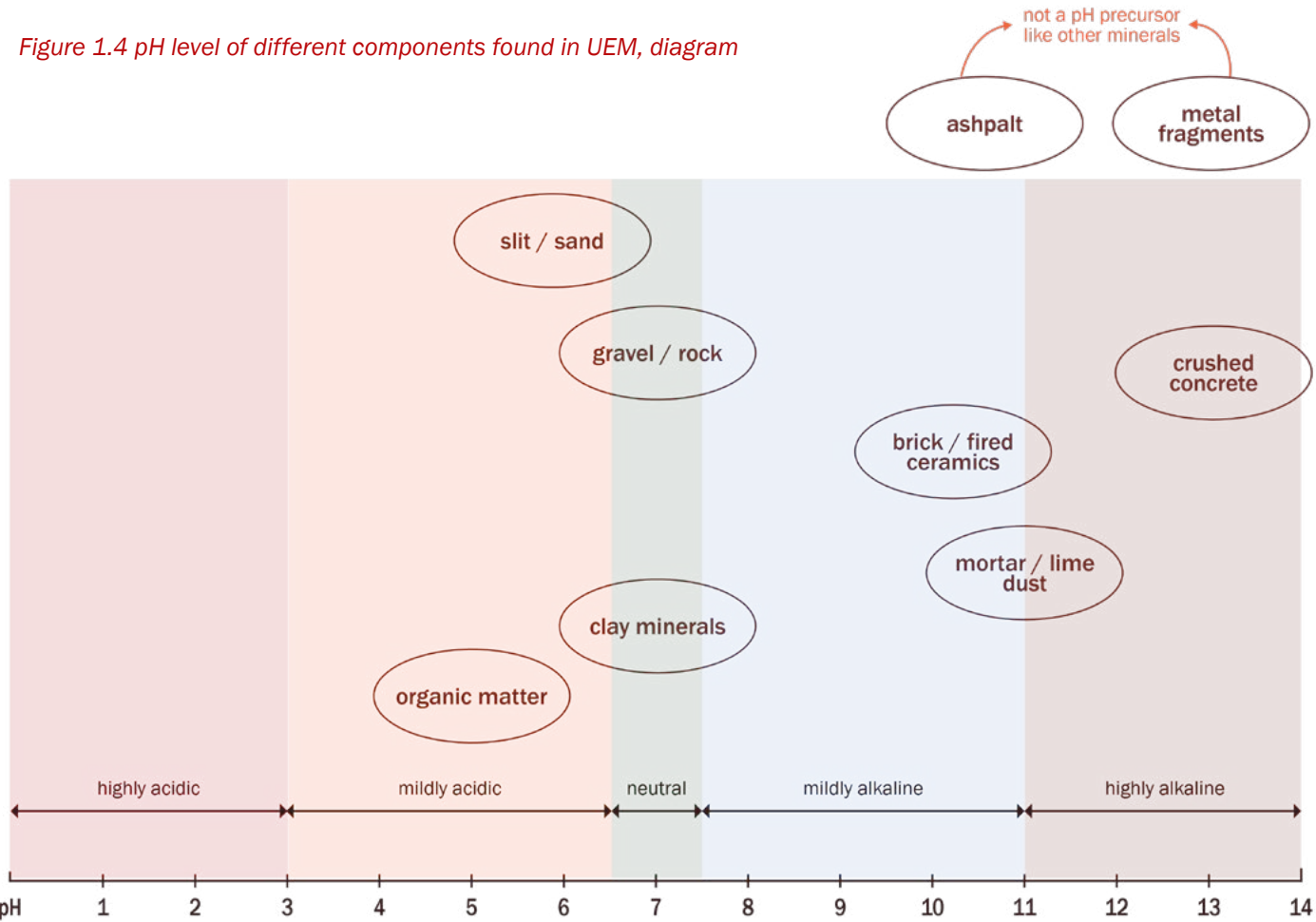
Figure 1.3 AAMs typology flowchart, diagram



Depending on the reaction mechanism and precursor chemistry, AAMs generally fall under two main categories: high-calcium systems with C-A-S-H gels (calcium-alumino-silicate-hydrate) as the main polymerization product and low-calcium systems with N-A-S-H gels (sodium-alumino-silicate-hydrate) as the main polymerization product, commonly known as geopolymers (Qin et al., 2022). See Figure 1.3 for the properties that each system structure leads to in the final cured AAM. There is also a hybrid system that combines both C-A-S-H gels and N-A-S-H gels within the structure, where the dominance of one or the other can either strengthen structurally or make it more resistant to aggressive environments (see Figure 1.3) (Qin et al., 2022).

When UEM is used as a precursor in an alkali-activated system, its reaction products depend on the chemical composition of the specific material sample. UEM containing calcium-rich cementitious particles may contribute to the formation of C-A-S-H phases, while reactive silica and alumina may support the formation of N-A-S-H phases. Depending on the precursor composition, hybrid reaction products may therefore be possible. There is sodium oxide content present consistently across most construction materials (those found in CDW) which reiterates the statement of N-A-S-H gel structures being created in the reaction using UEM (Marín-Cortés et al., 2024). However, confirming the resulting gel structure would require further chemical and microstructural characterisation.

Figure 1.4 pH level of different components found in UEM, diagram



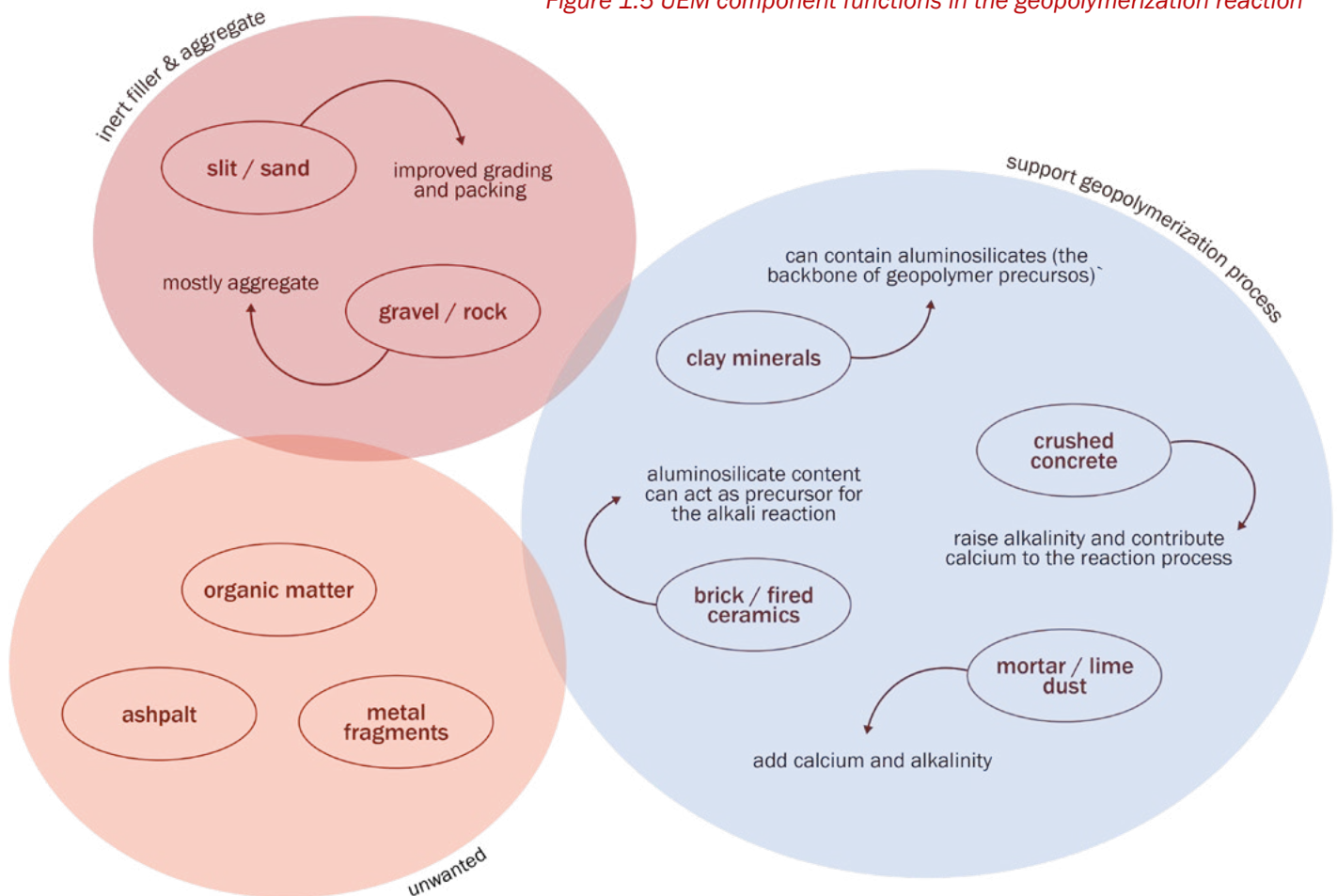
Previous studies have shown potential for UEM to be a geopolymerization precursor, wherein the main challenge is to increase its reactivity (Zhang et al., 2024). However, given the direction of this thesis research, the focus is on the use of a heterogeneous material in a digital fabrication system rather than engaging with the more detailed analysis of the chemical nuances. Based on existing and ongoing research, as well as the characterisation of the material components found in UEM, as an aggregate mix, its pH is more alkaline, which

reinforces its suitability to be geopolymerized (see Figure 1.4) (Marín-Cortés et al., 2024; Zhang et al., 2024). Figure 1.5 illustrates the different functions each material offers in the geopolymerization reaction by dividing them into two main functional groups, where one is acting as a filler in the overall matrix and the other helps support the reaction process.

Furthermore, AAMs are an active area of investigation amongst researchers for various reasons. As mentioned previously, some

geopolymer formulations have shown similar compressive capabilities to traditional cement. Additionally, AAMs can use industrial byproducts and even waste streams as feedstock to be created (as this thesis is exploring), which can help extend the useful life of these materials that would otherwise be discarded. More importantly, through different life cycle assessments, AAMs “are recognized as potential alternatives to ordinary Portland cement (OPC) to limit carbon dioxide emissions and beneficiate several wastes into useful products” (Qin et

Figure 1.5 UEM component functions in the geopolymerization reaction



al., 2022, p. 1). Komkova and Habert (2023) found that AAMs can produce up to 57% fewer CO₂-equivalent emissions than Portland cement concrete when variability in precursor and activator production is considered. It is important, however, to note that while certain studies do show a more sustainable outcome, certain material components give a more nuanced result given that “some studies (...) show that a part of geopolymer concrete has similar or even higher environmental impacts than OPC due to the negative environmental impacts brought

by sodium silicate production” (Qin et al., 2022, p. 2). As one of the most common parts of the alkali activator, it does impact the carbon footprint quite a bit. That is not to say that there are no options moving forward; currently, researchers are looking at alternative alkali activators from agricultural or industrial byproducts (Adesanya et al., 2021). Nevertheless, moving forward, the experiments and prototypes developed in this thesis use established geopolymer formulations as their basis.

02

material research

This chapter introduces the material exploration carried out to understand the processes of developing a geopolymer using urban excavation material (UEM) as the main precursor. Building upon established materials science research, this chapter is centred on understanding the material behaviour of the developed recipes. It starts with a series of locally excavated soil samples to establish the contribution of the cement-based and ceramic-based aggregates that are found in excavation sites. Afterwards, the experiments become more focused on a single precursor sample to find even more nuances in the reactions. The last stage of the material research done explores incorporating other agents in the matrix to alter the density of the final cured geopolymer.

Figure 2.1 excavation material samples:
site A (top), site B (bottom)

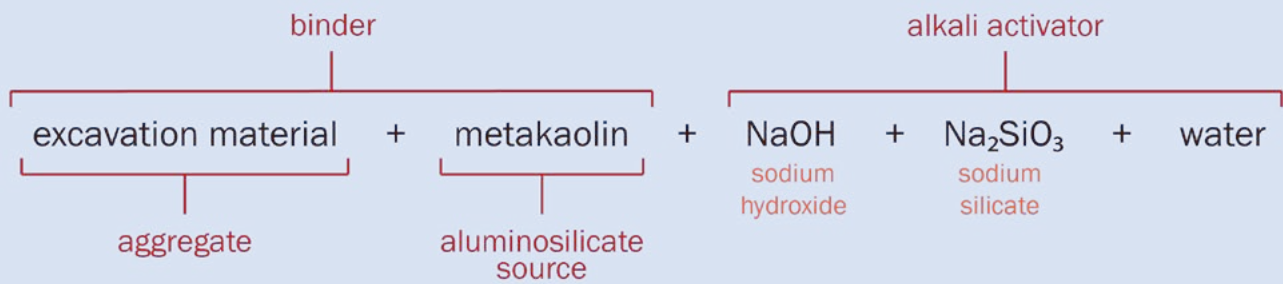


Geopolymerizing Urban Excavation Waste Establishment of UEM as Reaction Precursor

The material extracted from the ground in urban areas is all different. Some sites present higher concentrations of anthropogenic material, others have more organic content. Figure 2.1 shows photographs taken of the two samples that were used in the first material experimentation. Sample A was extracted from an earth construction site in the Valldaura Labs near the municipality of Cerdanyola in Barcelona, Spain. Sample B was extracted from a roadside excavation site in the neighborhood of Poblenou in Barcelona, Spain. It is important to note that even though Sample A is mostly organic soil, which differs completely from the target precursor of this research, the purpose of this first experiment is to validate the suitability of UEM as a geopolymer precursor. Therefore, it acts as a control and reference rather than a representative UEM.

As documented in the literature, the chemical composition that enables the reaction of geopolymerization is the combination of an aluminosilicate precursor and an alkali activator. Previous geopolymer research has explored different precursors to then geopolymerize, meaning that in many cases an aluminosilicate source has been added to the recipes to contribute to more dissolved Si (silicon) and Al (aluminium), creating more gel structures. These then fill in the voids and bind the less reactive aggregate in the precursor creating an overall homogeneous matrix (Provis, 2018; Joseph Davidovits, 2020). With this said, the first set of material experiments was to test how different concentrations of added aluminosilicate source would affect the strength of the cured mix. The chosen aluminosilicate source for this thesis was metakaolin, which is calcined kaolinite clay making it a highly reactive pozzolan (Panesar, 2019). One of the reasons to use metakaolin is because its “favorable ratio of silicon to aluminum [makes it] an ideal solid for the synthesis of geopolymers” (Vogt et al., 2019, p. 1). Moreover, metakaolin is an aluminosilicate source that when activated can harden at room temperature without the need for additional heat (Vogt et al., 2019), which was crucial to the fabrication conditions throughout the research done for this thesis.

recipe components



20%

MATERIAL PART

precursor	1.27
aluminosilicate	0.20
alkali activator	0.56

sample A

density (kg/cm³) 1939
stress (mPa) 4.234



sample B

density (kg/cm³) 1886
stress (mPa) 2.855



30%

MATERIAL PART

precursor	1.07
aluminosilicate	0.30
alkali activator	0.52

density (kg/cm³) 1924
stress (mPa) 5.277



density (kg/cm³) 1907
stress (mPa) 7.980



40%

MATERIAL PART

precursor	0.94
aluminosilicate	0.40
alkali activator	0.51

density (kg/cm³) 1866
stress (mPa) 7.944



density (kg/cm³) 1848
stress (mPa) 10.49



As a material, similar to concrete, that exhibits substantially greater strength in compression than in tension (Provis and van Deventer, 2009; Yankelevsky, 2024), the sample prototypes were designed specifically to be tested using a hydraulic press. Each geopolymer recipe was cast into a 10 x 10 x 10 centimetre cube (see layout on page 20). Experiment 01 tested 3 recipes using 2 different precursor samples, Sample A and B, as introduced earlier. The methodology to make the samples was as follows. First, the alkaline activator was prepared by creating NaOH solution (sodium hydroxide solution), mixing NaOH pellets with water; after this solution cooled to room temperature, it was added to sodium silicate. This mixture was then left to cool again and then mixed into the solid materials: the precursor with added metakaolin. After combining into a homogeneous mixture, it was poured into the cube mould, covered with plastic wrap and left to cure.

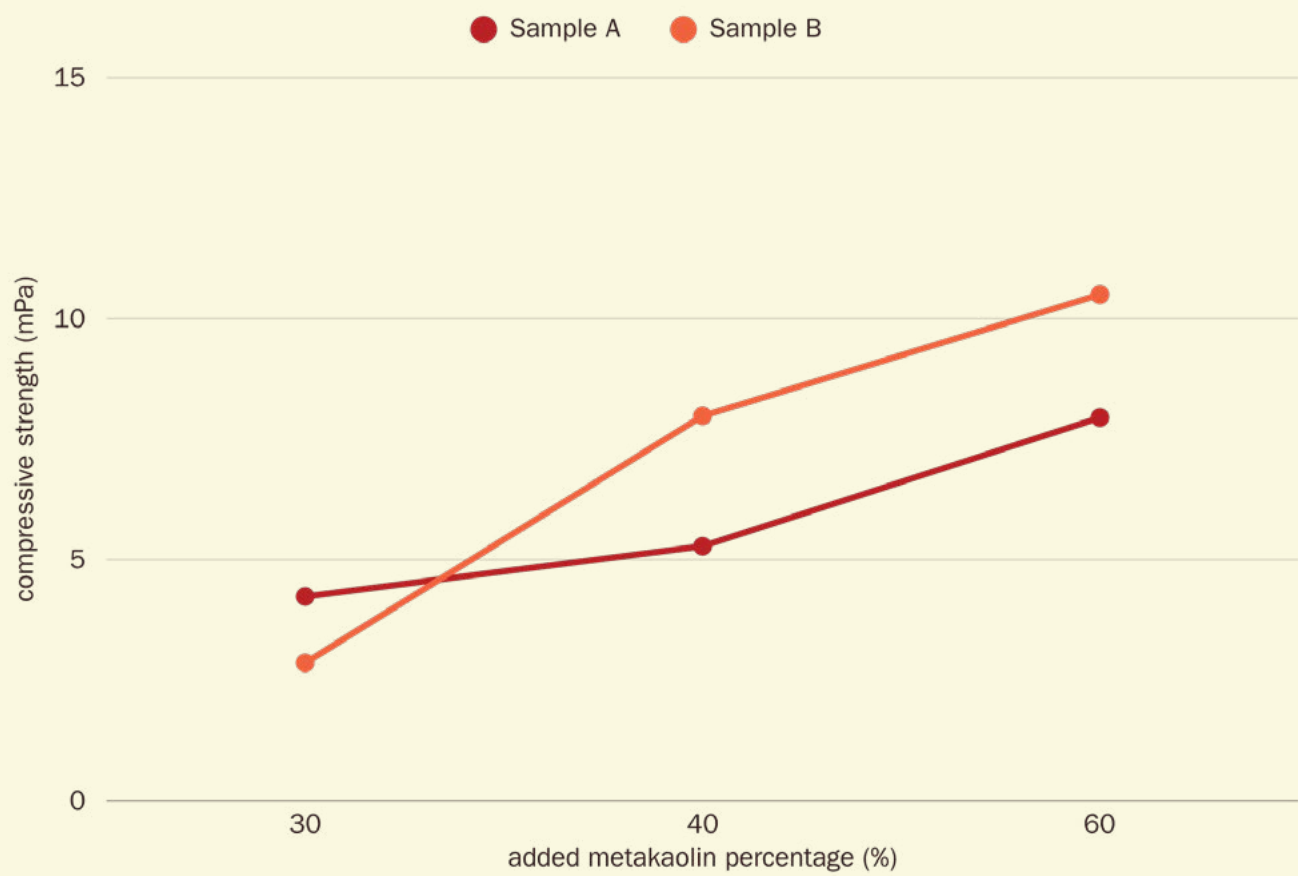


Figure 2.2 Compression test results of Experiment 01, graph

base

MATERIAL	PART
precursor	1.30
aluminosilicate	–
alkali activator	0.55

density (kg/cm³) 1589
stress (mPa) 1.488



Based on the results, Experiment 01 indicates a general positive relationship between added metakaolin content and measured compressive strength, although the reduction observed in the 30% formulation shows that the relationship was not entirely linear (see Figure 2.2). The higher the percentage of metakaolin in the mix, the greater its measured compressive strength. These results align with the previous research gathered, where the higher the concentration of Si and Al the more it can react with the alkaline creating the bonds that strengthen the material. One key observation from this experiment, however, is that despite the drop in the 30 percent prototype, Sample B shows a higher compressive strength overall. This further justifies the suitability of UEM to be a geopolymer precursor. Another important note for this first experiment is that the curing time for the samples was 14 days only; there is literature review that has stated through different experiments that extending the curing time enhances the strength of the geopolymer (Nurrudin et al., 2018).

10%

MATERIAL	PART
precursor	1.33
aluminosilicate	0.10
alkali activator	0.60

density (kg/cm³) 1708
stress (mPa) 6.613



15%

MATERIAL	PART
precursor	1.35
aluminosilicate	0.15
alkali activator	0.63

density (kg/cm³) 1724
stress (mPa) 9.229



20%

MATERIAL	PART
precursor	1.36
aluminosilicate	0.20
alkali activator	0.66

density (kg/cm³) 1618
stress (mPa) 12.81



Compressive Strength

After finalizing this initial experiment, the testing was focused solely on using UEM as the main precursor. In Experiment 02, a second set of prototypes was made with Sample B to test the compressive strength of the recipe, maintaining different percentages of added metakaolin (aluminosilicate source), but at a lower quantity (see layout on page 23). For this set, a “base” recipe was tested, where no metakaolin was added to the mix, and it was solely the alkaline solution. For this set of prototype samples, the same methodology from Experiment 01 was used.

Following a curing period of 21 days (a week longer than in Experiment 01), the compression test results were as seen on Figure 2.3. Similar to the results from Experiment 01, this set shows a direct relationship between the compressive strength and the added aluminosilicate percentage. The trendline illustrates the general tendency within the tested range; it should not be interpreted as predicting the performance of formulations with higher metakaolin contents. An interesting observation of both experiments is that the highest percentage used in Experiment 01 (which was of 40%) is higher than in Experiment 02 (being of 20%), yet the compressive strength for the latter surpassed that of the first one, which can be related to the curing time and process. There was higher rigour in the second set to cure the samples for longer and with proper covering. These are all possible contributing factors to why Experiments 01 and 02 varied as they did.

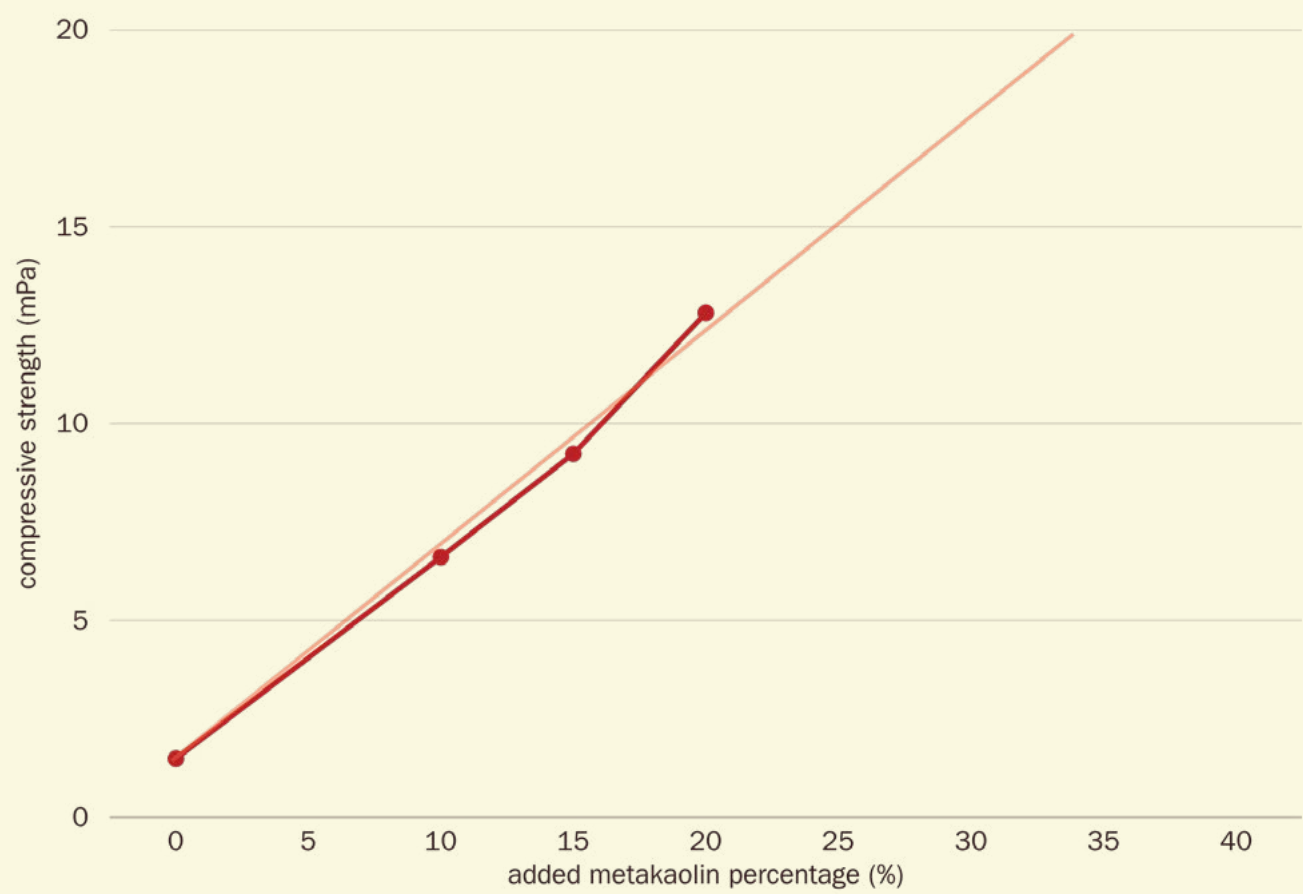


Figure 2.3 Compression test results of Experiment 02, graph

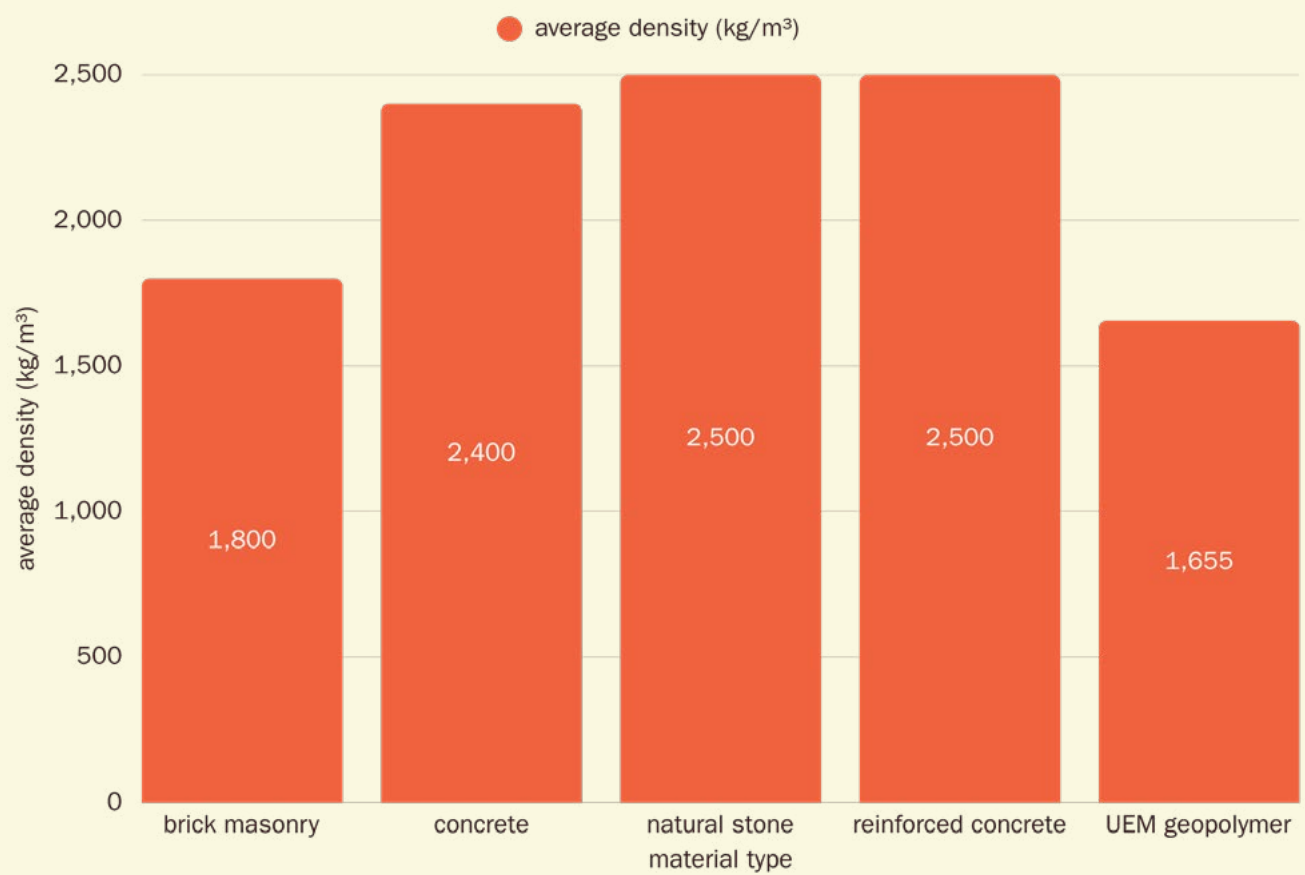


Figure 2.4 Densities of common construction materials and UEM geopolymer, chart | Source: (Malakatas, 2022)

Material Density

During the process of developing the geopolymer recipes and the sample cubes, one property that stood out was the weight. The average density of the UEM geopolymer was comparable to that of a brick. When discussing the properties of a construction material, typically “materials with higher mechanical characteristics [result] in aggregates with higher density values” (Silva, de Brito and Dhir, 2014, p. 206). The general explanation for this is that higher density materials are less porous meaning that they absorb less water and thus have better resistance to environmental exposure. Although density can influence porosity and water absorption, these relationships were not tested directly in this study. Here, density was considered primarily in relation to component weight and the feasibility of architectural-scale fabrication. So now the compression test results and the density of the geopolymer have shown that there is buildability potential. Nonetheless, this does not negate that it is still a heavy material, and with the goal to implement it with digital fabrication to achieve architectural scale components it does raise the question: **could it be made lighter?**

Using insights from the Digital Matter Research Studio at IAAC and previous studies, this thesis’ material research briefly shifted towards finding a strategy to lighten the geopolymer. Going off of the logic from the peer reviewed papers presented earlier, the goal was to create more porosity in the mix to achieve a lower density. The approach that was explored in this research used a “foaming system” to incorporate into the geopolymer recipe (see page 24). The idea is to use the combination of a foaming agent stabilised by a surfactant to create air bubbles in the mix, consequently lowering the density and thus the weight of the cured geopolymer. Following a similar approach to Kim et al. (2022, p. 4), Experiment 03 “used hydrogen peroxide (H_2O_2) to derive the [geopolymer] foam structure”. The reason for this is to allow for the curing to be done at room temperature while preserving the foaming structure generated from the reaction (Kim et al., 2022). The mechanism behind this reaction is that hydrogen peroxide’s innate instability as a compound is destabilised even more when it comes in

contact with the alkalinity of the geopolymer mix; it accelerates its decomposition separating the compound into water and oxygen molecules. As a gas, the oxygen creates these air bubbles in the mix that are stabilised by the surfactant which lower the surface tension of the air pockets (Kim et al., 2022). In contrast to Kim et al. (2022), this thesis uses polysorbate 80 as the surfactant, instead of sodium dodecyl sulfate, because it is more readily accessible and commercially available.

To explore a variety of densities, Experiment 03 uses different concentrations of hydrogen peroxide to achieve three different levels of porosity hereafter denoted as “light foam”, “mid foam” and “high foam”, where each consecutively has more hydrogen peroxide (see layout on page 24). The same methodology as the previous two experiments was implemented (Experiments 01 and 02) to make the geopolymer recipe. The additional steps to create the foaming effect was to first add the surfactant (polysorbate 80), mix the mixture thoroughly and then right before casting into the same cube moulds, the foaming agent (hydrogen peroxide) is added and stirred between 30 to 60 seconds. To take into account the expansion of the mix as it foamed, the moulds were not filled completely, leaving room for the geopolymer to rise and then set. An AI-assisted preliminary calculation was used to estimate the initial filling volumes based on the dimensions of the moulds and assumed expansion rates. These estimates served only as a starting point and were subsequently adjusted through direct observation of the material trials. As expected, there was a distinct variation in the expansion rates for the three concentrations, although not drastic enough to where the cast mixture amounts varied significantly. It was observed that pouring the mixture about approximately 60–75% of the mould volume was enough to allow the geopolymer to expand to fill up the mould, after covering the top.

**geopolymer
base recipe**

+

foaming system

polysorbate 80 + hydrogen peroxide

surfactant

foaming agent

low foam

MATERIAL	PART
excavation m.	1.0000
aluminosilicate	0.1000
alkali activator	0.4200
polysorbate 80	0.0035
hyd. peroxide	0.0055

density (kg/cm³) 1411



mid foam

MATERIAL	PART
excavation m.	1.0000
aluminosilicate	0.1000
alkali activator	0.4200
polysorbate 80	0.0035
hyd. peroxide	0.0110

density (kg/cm³) 1211



high foam

MATERIAL	PART
excavation m.	1.0000
aluminosilicate	0.1000
alkali activator	0.4200
polysorbate 80	0.0035
hyd. peroxide	0.0187

density (kg/cm³) 1025



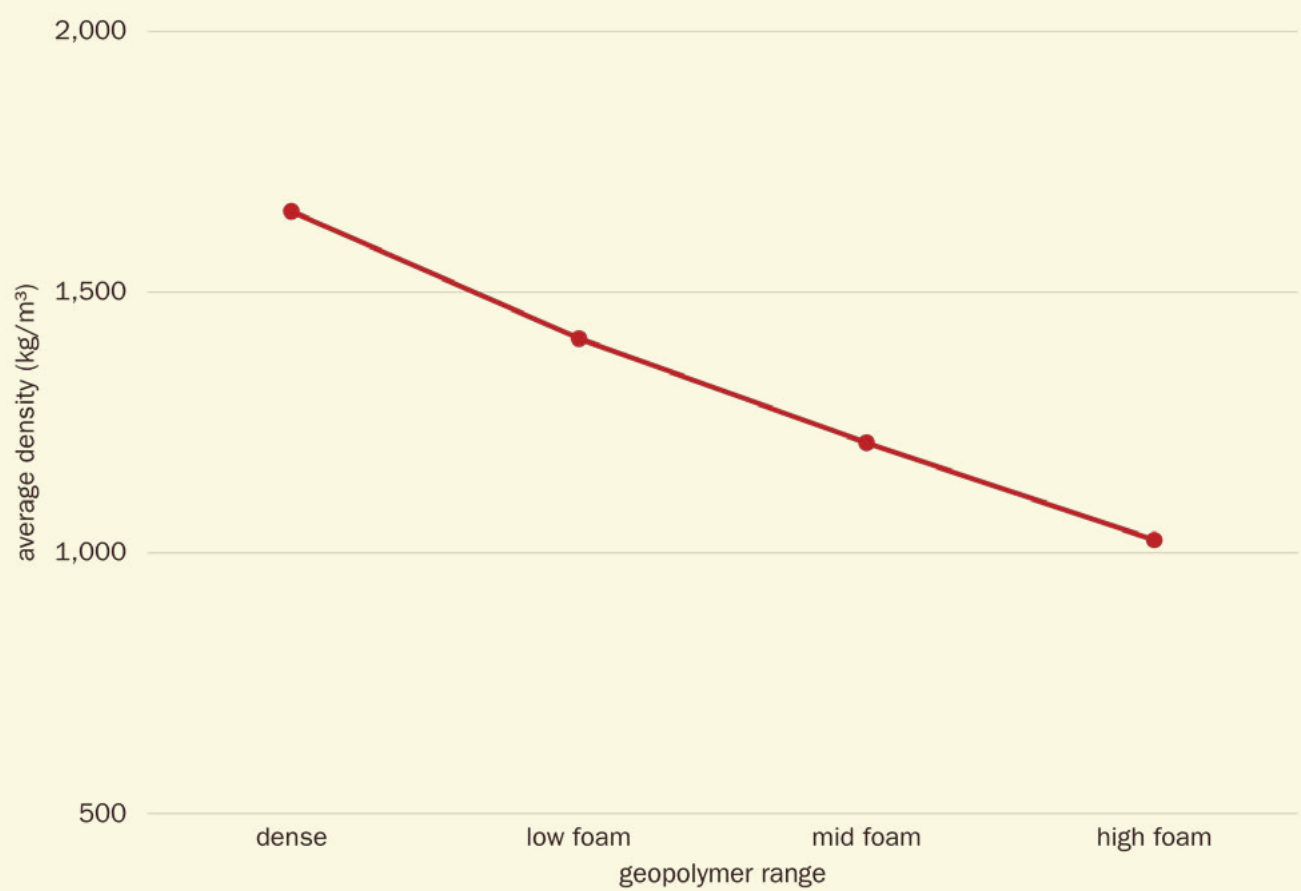


Figure 2.5 Average density results from Experiment 03, graph

After curing for three weeks, the samples were demoulded and weighed. Their densities were calculated by dividing the mass of each sample by the fixed internal volume of the cube mould. All the samples were made in moulds of the same volume and after being poured, every mould was covered with plastic to control the expansion. In order to compare these foamed samples with an ordinary UEM geopolymer, the average density of the samples from Experiment 02 was calculated. As expected, there is a visible decline in the density of the material the higher the foaming level, which as stated previously is associated with the concentration of hydrogen peroxide added (see Figure 2.5).

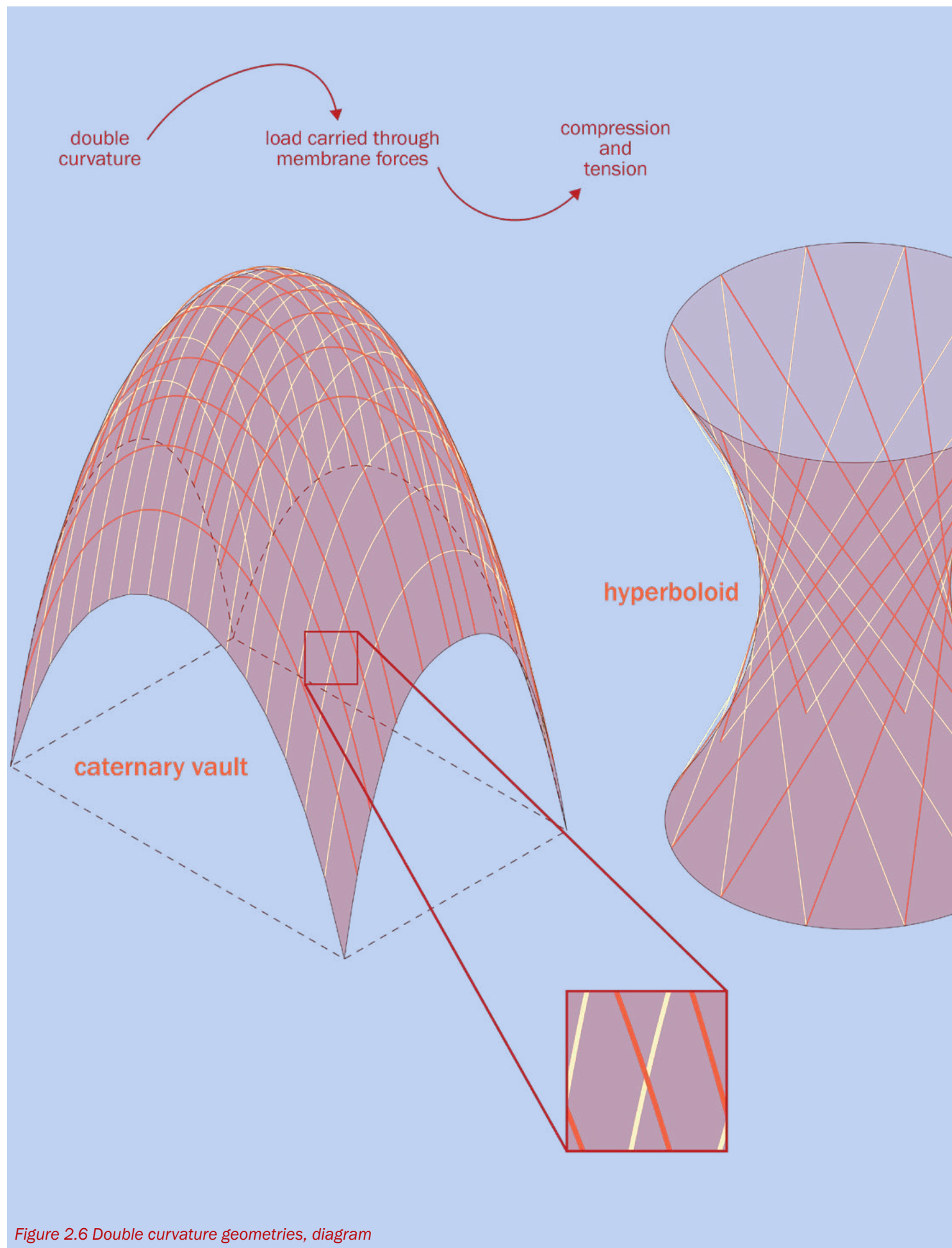
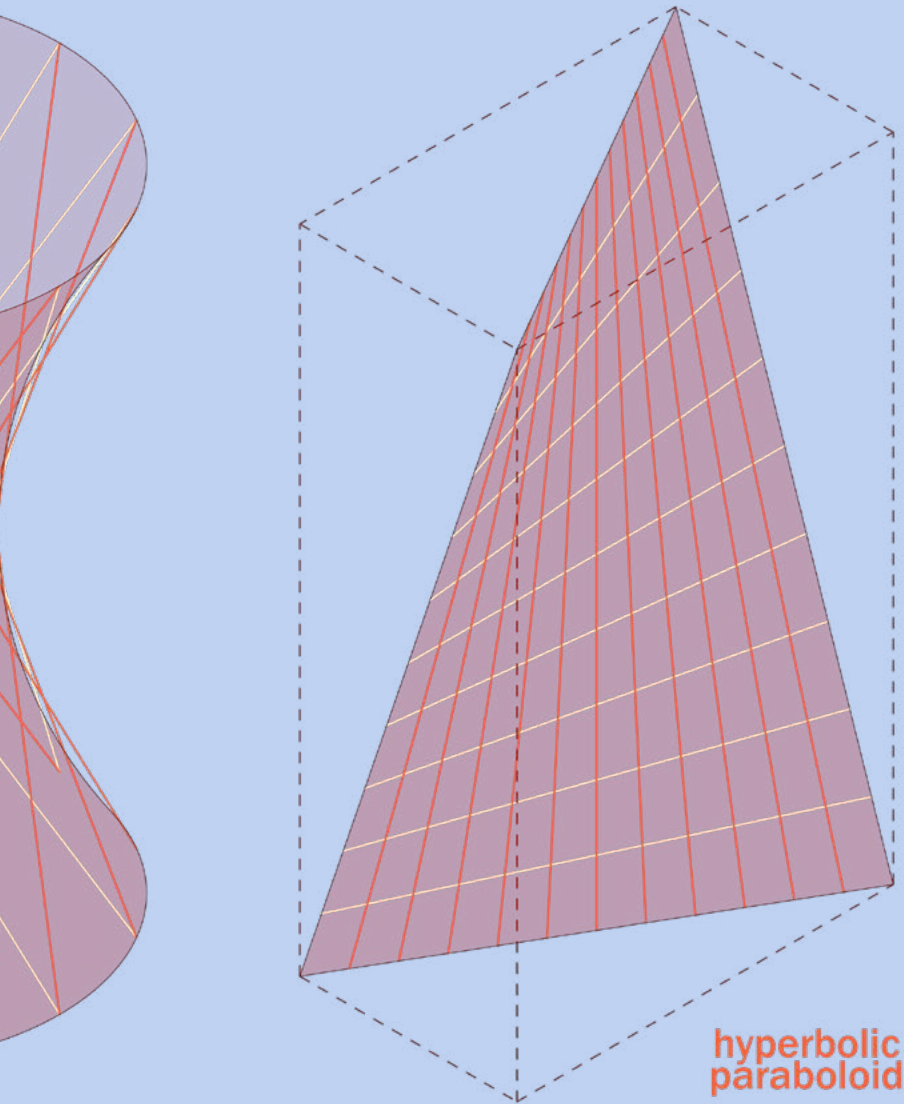


Figure 2.6 Double curvature geometries, diagram



Geometric Logic

The ambition behind this research, however, goes beyond the conventional practices. In the current discourse, researchers in the construction and architecture industry are bringing up the idea of material efficiency. More importantly, the discussion revolves around how material efficiency is rarely the primary driver for building design, and decisions are often made around constructability, cost, and conventional practice (Dunant et al., 2021). Although Dunant et al. (2021) discusses this idea from an engineering perspective and talks about the structural framework of a building, it directed the investigation towards the form and geometry that would best suit this material.

In response to the findings on this geopolymer working mostly in compression, the goal was to explore the geometries that would be most suitable for optimising the structural performance of the geopolymer. Geometry studies have shown that ruled surfaces are quite suitable for materials that are compression dominant; in their study Li et al. (2024) refer to them as compression-dominant structures and specifically talk about materials like concrete and stone being effective for these geometries. Ruled surfaces are defined as surfaces that have been generated by moving a straight line through space along one or more guiding curves. From an architectural perspective, they are a favourable geometry as they are easy to fabricate

(Velimirović et al., 2009). Ruled surfaces can be created from flat materials, making them easy and fast to fabricate through efficient manufacturing workflows (Baharami et al., 2025).

Other studies show that doubly curved surfaces are another successful geometry to implement in architecture (see Figure 2.6). In terms of structural optimisation, doubly curved surfaces can achieve increased stiffness by distributing membrane forces in two principal directions (Cao, Kotnik and Schwartz, 2023). As the forces are acting through compression and tension (membrane forces), bending is reduced. Moreover, in their review discussing shell structures as an example, Veres and Tănase (2025, p. 1) state that “distributing loads primarily through membrane forces across a curved surface, allows for the creation of expansive structures with a minimal expenditure of material”.

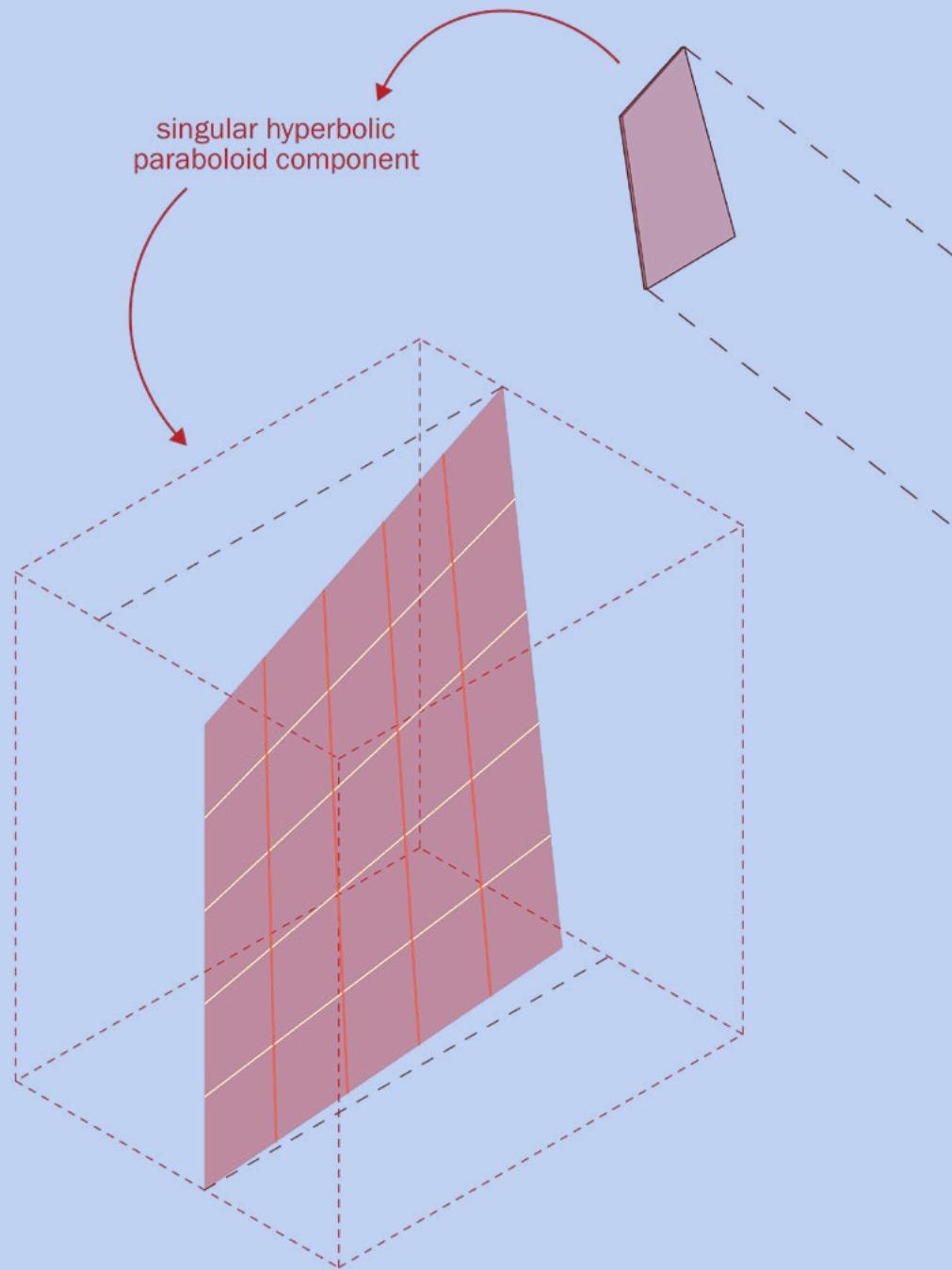
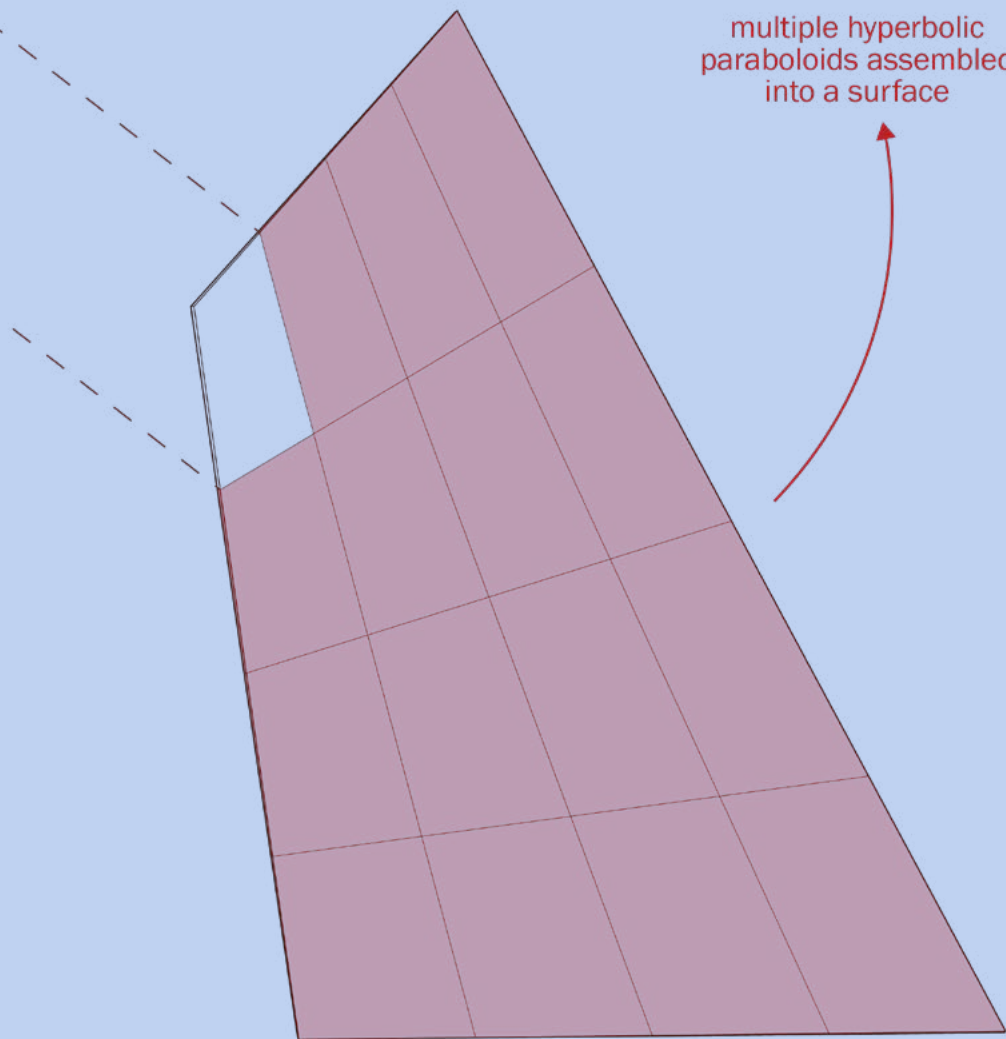


Figure 2.7 Hyperbolic paraboloid part of a modular structure, diagram



Hyperbolic Paraboloid

Considering the variety in double curvature geometries, this research focuses on the hyperbolic paraboloid. This is also known as the saddle-shaped surface, defined as a quadratic surface of double curvature, that curves upward along one principal direction and downward along the other (Weisstein, N/A). This shape was chosen for this thesis, not only because of its structural efficiency, but also because of its potential as a modular component. In their paper, Cao et al. (2023) discuss the characteristics of the hyperbolic paraboloid (known as “hypar”) which make it a versatile geometry to design architecture with. As an individual shape it possesses all the characteristics mentioned above regarding ruled surfaces and double curvatures, but as a modular component, where multiple hypars are aligned and assembled together, the “smooth poly-hypar surfaces are ruled locally, while globally appearing to be continuous freeform” (Cao, Kotnik and Schwartz, 2023, p. 2). More importantly, the assembled structure remains structurally sound as the “double curvature of the individual hypar modules and the smooth connections (...) between them ensure global bending-free structural behavior” (Cao, Kotnik and Schwartz, 2023, p. 2).

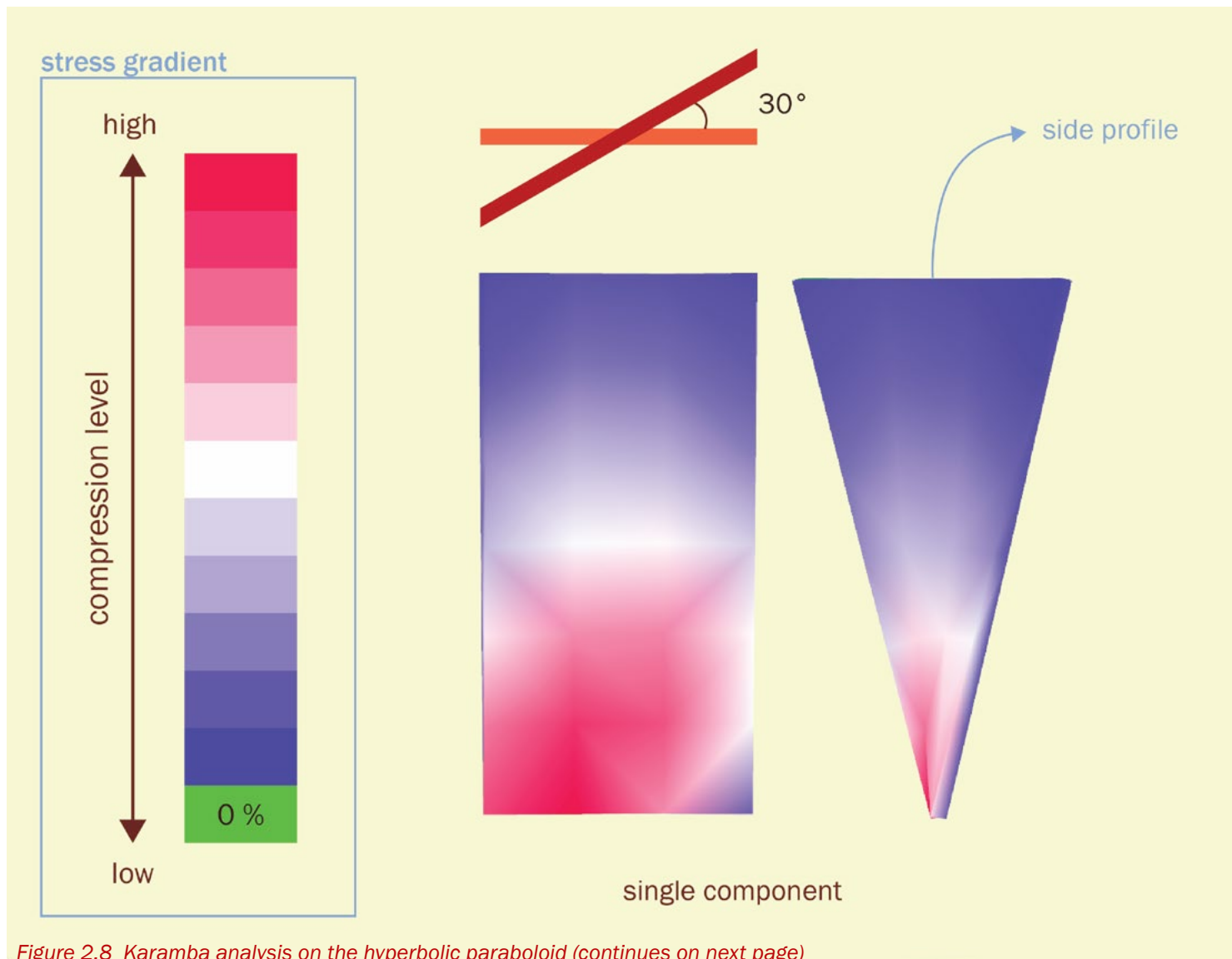
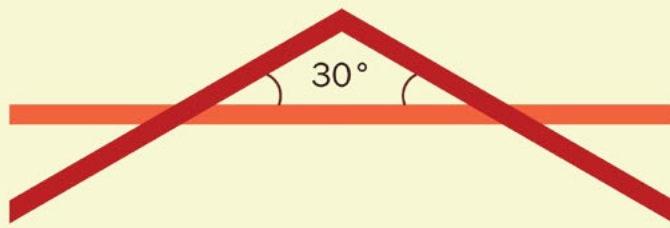


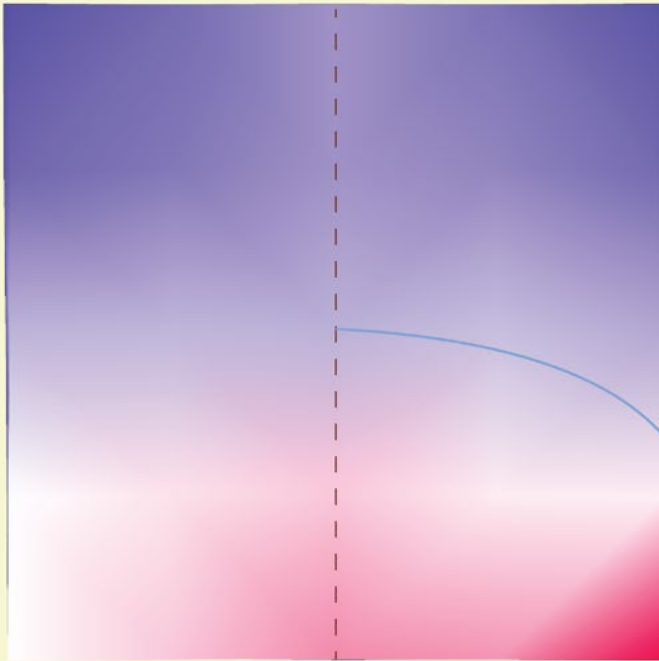
Figure 2.8 Karamba analysis on the hyperbolic paraboloid (continues on next page)

Functional Gradient

Following the research on geometry and material efficiency, a structural analysis was conducted on the hyperbolic paraboloid. Using the Karamba plugin in Grasshopper, a vertical standing hypar was analysed (see Figure 2.8). The way the simulation was set up is as follows: the hypar was supported from the bottom mainly from the two corner points. The loads applied to the structure were a uniform surface load applied downward from the top of the standing hypar and the gravity/self-weight of the piece itself. Then for the structure itself, it was assumed as a 1:1 scale piece, with a height of 1.0 metre, a width of 50 centimetres and a thickness of 4 centimetres; it was assigned the material properties of concrete as its the closest one in typology. As the legend scale in Figure 2.8 shows, the warmer (more red) the colour, the greater the compression. There was a clear pattern in the levels of compression across the analysed orientations after comparing different hyperbolic paraboloids, rotated at different angles, where the highest compressive stresses/forces occurred near the bottom of the hypar and the compression gradually decreased towards the upper portion of the component.



legend



each component is created
by rotating the top edge
from the center point at the
designated angle

component
aligned edge

two components

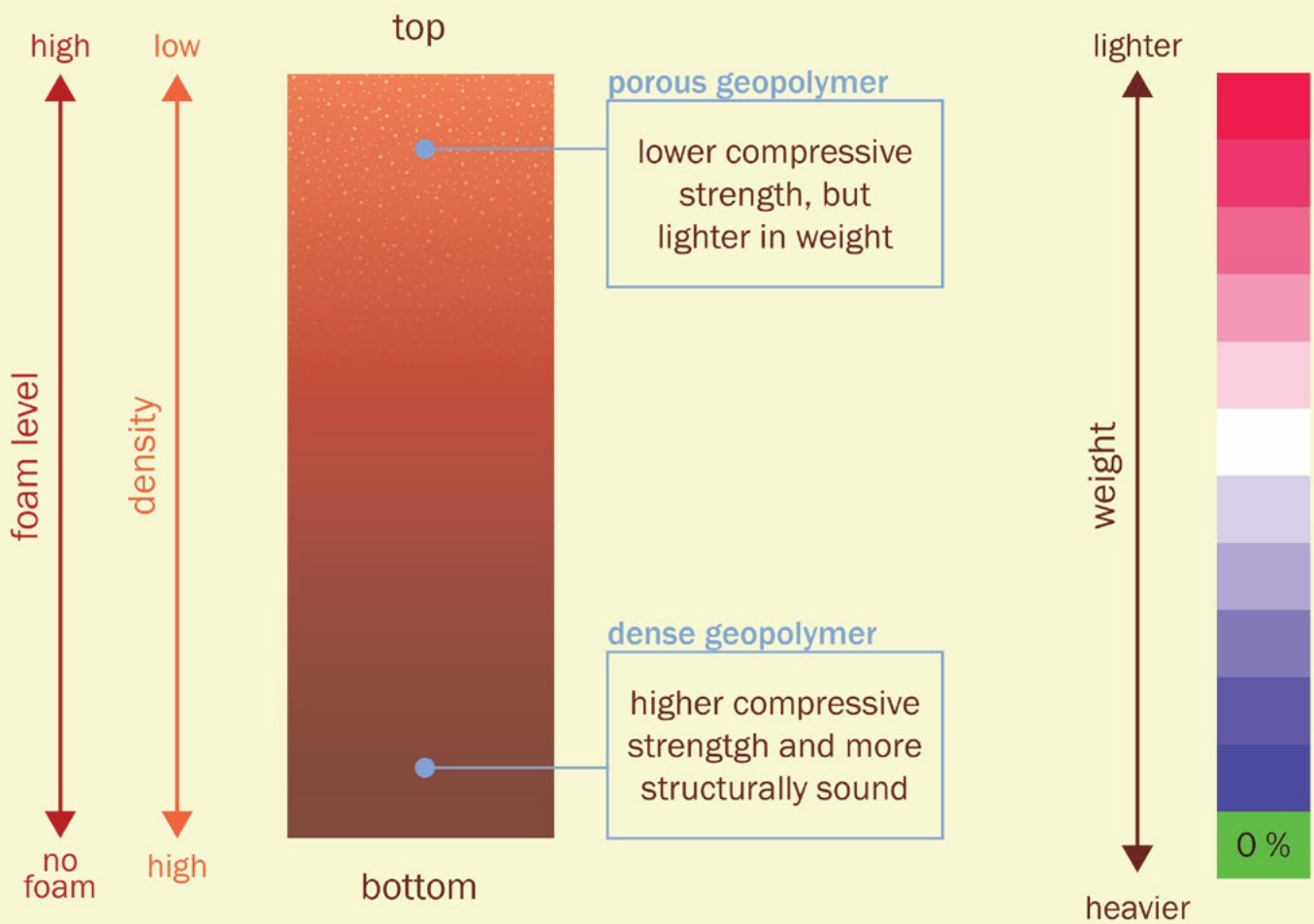
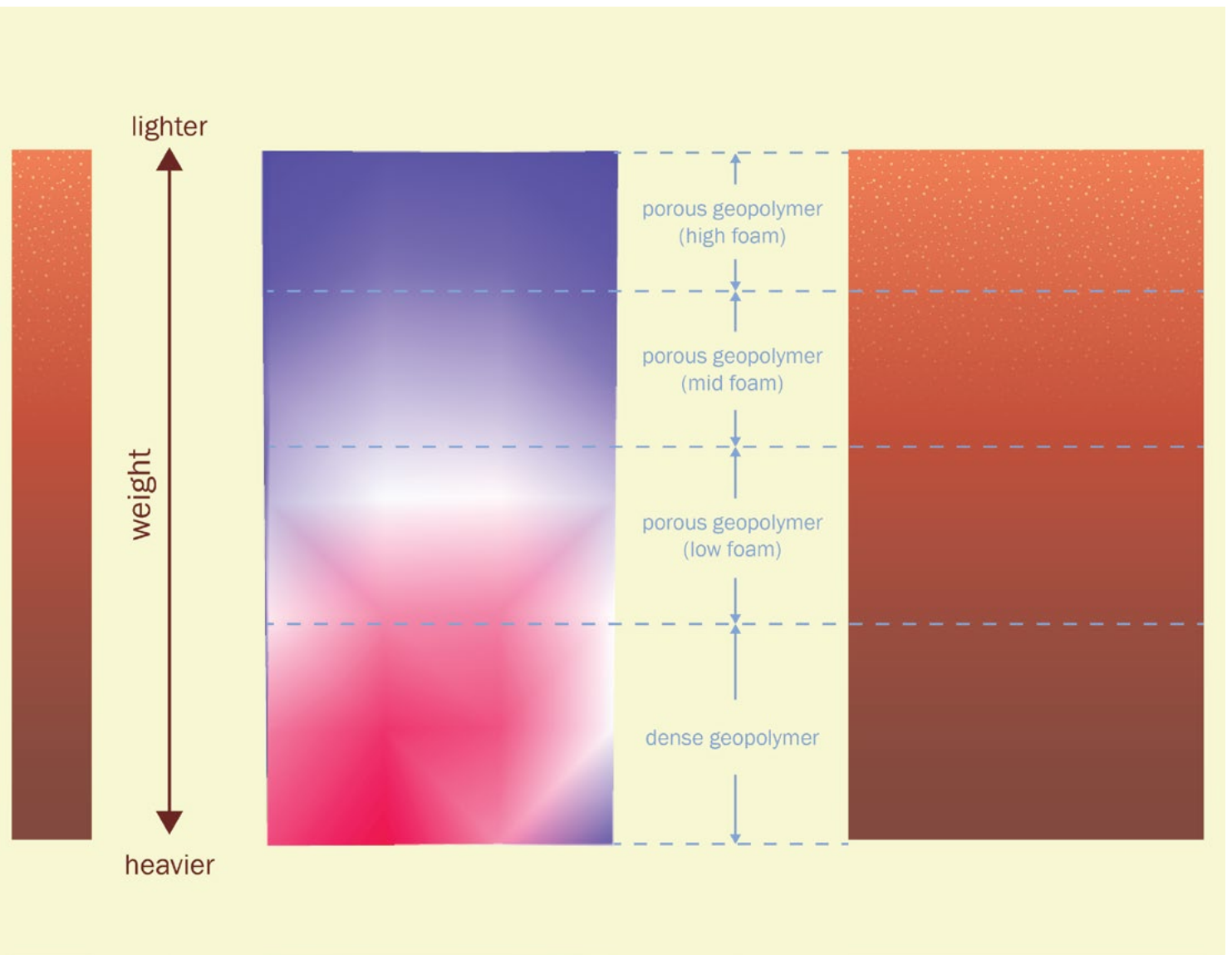


Figure 2.9 Functional gradient proposal based on Karamba analysis (continues on next page)

Based on all these findings, a material system was proposed (see Figure 2.9). The experiments demonstrated that metakaolin content can influence compressive strength, while the introduction of hydrogen peroxide can produce geopolymer samples with different densities. Together, these findings suggest the potential for a functionally graded material system. However, further testing is required to quantify the compressive performance of each foamed formulation. Compiling them into a gradient system offers the possibility of creating different variations of the geopolymer according to the requirements. With the development of this material gradient and the compression distribution of the hyperbolic paraboloid, the next step was to combine this knowledge into a functionally graded material system. The proposed distribution therefore places the denser geopolymer formulation in the zones experiencing greater compression and progressively lighter formulations towards the upper portion of the component. This arrangement constitutes a design hypothesis based on the observed density variations and structural analysis; further testing is required to verify the compressive performance of the complete graded component.



Material Research Limitations

The material experiments conducted in this thesis represent a preliminary investigation into the use of urban excavation material as a precursor for alkali-activated construction materials. Due to time and resource constraints, the experiments were limited to three excavation material samples and a restricted number of specimens per formulation. Differences in precursor composition, curing duration and curing conditions also mean that results from separate experiments cannot be directly compared without further controlled testing. Moreover, while the experiments demonstrated that metakaolin content influenced compressive strength and that foaming altered material density, the compressive performance of the foamed formulations was not tested. Long-term durability, tensile performance and detailed chemical and microstructural characterisation were also beyond the scope of this research. Consequently, the results establish the feasibility and design potential of the proposed material system rather than a fully validated construction material.

Research Question

Collectively, this material investigation was done to achieve the original research objective of using UEM as a material resource in a digital fabrication system. After discarding extrusion as the main fabrication strategy, the investigation was directed towards other robotic fabrication systems. Combining all these ideas and data prompted the research question:

How can robotic fabrication enable the production of functionally graded, urban-waste-based geopolymers?

03

dynamic casting system development

In this chapter, the robotic fabrication system that was designed is introduced, beginning with the initial hypothesis and continuing through its development, testing and final configuration. After looking at the state of the art, there are three precedents that stood out, from which this thesis extracted key ideas. The process of designing the system began with a series of tests to determine the materials that would be used to make up the physical setup. Subsequently, a series of trials were conducted with the robotic arm to test the fabrication system before finalising it.



hypothesis This investigation is guided by the assumption that urban excavation waste's heterogeneity can be leveraged through functionally graded geopolymers, cast with a robotic system without the need for extrusion.

State of the Art

In this paper, Márquez-Muñoz (2023) showcases how a group of students experimented with fabric formwork inspired by Miguel Fisac. In this exploration, there is a clear demonstration of the behaviour of concrete as a cast material against phenomena like hydrostatic pressure and textile deformation. There is clarity in the idea that the material itself has a clear role in the design process as there is a constant participation between the fabric formwork and the concrete (in this case). Another important point made in this paper is the inexpensiveness of fabric and its potential for reuse, which make it an even more attractive material to implement. Overall, they have demonstrated the architectural potential of flexible fabric formwork through educational experimentation.

More on the topic of geometry, Scheder-Bieschin et al. (2024) offers a comprehensive review and classification of bending-active formwork systems from historical precedents to current research. Although they approach formwork from a different strategy, in their work they emphasize the integrated design of geometry, fabrication and construction. Despite the focus on bending structures, the case studies demonstrate that structural assessment and optimisation favour shell-like geometries as the most structurally efficient configurations.

The final contribution to this review is from Yang et al. (2019), in which they developed a robotic, reconfigurable fabric formwork capable of producing multiple doubly-curved panels from a single mould. As the closest precedent to this thesis' work, they fabricated hyperbolic paraboloids using fabric formwork that is shaped with the help of a robotic arm. More importantly, the fabrication system they develop offers formwork that is reusable and can be adapted to different dimensions and thus create variations of the hypar. The use of the robotic arm is also

key in being able to have precise control over the movement of the fabric to achieve the desired parameters on the hyperbolic paraboloid.

The research presented in earlier chapters demonstrate developments in alkali-activated materials and their potential as low carbon alternatives when waste streams are used as feedstocks. In parallel, the reviewed precedents demonstrate advances in flexible and reconfigurable formwork for geometrically efficient structures. However, these systems focus primarily on geometric and formwork adaptability while using relatively homogeneous conventional materials. The integration of functionally graded waste-based materials within robotic formwork systems remains largely unexplored. This thesis addresses that gap by combining robotic formwork transformation with the staged casting of dense and foamed UEM geopolymer formulations to produce modular hyperbolic-paraboloid components.

See the table on page 37 for the overall analysis of the state of the art.

Dynamic Casting

In this thesis, dynamic casting refers to the staged casting of material into a flexible formwork that is actively reconfigured by a robotic arm. This differs from established dynamic casting processes in which a rigid formwork moves continuously relative to setting concrete.

The system consists of the continuous movement of the formwork as the material is being cast into it. It is dynamic because more than the formwork attributing the geometry to the cast material, it is the movement from start to finish that distributes the material and shapes the final geometry. To create the double curvature geometry of the hyperbolic paraboloid, the robotic arm translates along the Z-axis while simultaneously rotating to a defined angle.

state of the art analysis

precedent	Márquez-Muñoz (2023), Casting Concrete with Fabric Formwork	Scheder-Bieschin, Van Mele and Block (2024), Bending-Active Formwork Systems for Concrete Shells	Yang, Loh and Leggett (2019), Robotic Variable Fabric Formwork	This thesis: robotic dynamic casting of UEM geopolymers
material	Conventional concrete	Primarily conventional concrete and related cast materials across the reviewed case studies	Conventional concrete	Alkali-activated urban excavation material with dense and foamed formulations
formwork strategy	Flexible textile formwork deforms in response to the pressure of the cast concrete	Flexible sheets or structural elements are elastically bent into curved formwork for shell construction	Fabric formwork is shaped by a robotic arm to produce doubly curved panels, including hyperbolic paraboloids	Flexible cotton formwork constrained by reusable steel rods and transformed during casting
role of digital fabrication	No robotic fabrication; the experiments focus on manual construction and material-formwork interaction	Digital modelling and fabrication support the prediction, manufacture and assembly of bending-active systems	The robot precisely positions and deforms the fabric formwork according to digitally defined geometric parameters	The robot simultaneously translates and rotates the upper frame, shaping the flexible formwork through staged casting operations
reconfigurable ?	Partially. The fabric can be reused, but the system is not digitally controlled to produce precise variations	Yes. Several systems can be adjusted or reused to produce geometric variation, depending on their construction	Yes. A single formwork system can produce panels with different dimensions and curvatures	Yes. The movement, frame dimensions and rotation can be digitally adjusted for different components
functionally graded ?	No	No	No	Yes, as a proposed density-gradient system
relevance	Demonstrates how concrete and flexible fabric can jointly determine geometry and surface expression. However, it does not incorporate robotic control, heterogeneous waste-based materials or functional grading.	Establishes the potential of adaptable formwork for materially efficient shell geometries emphasizing the integration of design, structural analysis and fabrication. It does not investigate active robotic transformation during casting or functionally graded waste-based material.	Closest fabrication precedent: it combines robotics, reusable fabric formwork and geometric variation. However, the formwork is configured for each geometry rather than continuously transformed as functionally graded material is cast in stages	Integrates heterogeneous urban-waste-based material, staged functional grading and robotic formwork transformation to produce modular hyperbolic-paraboloid components.

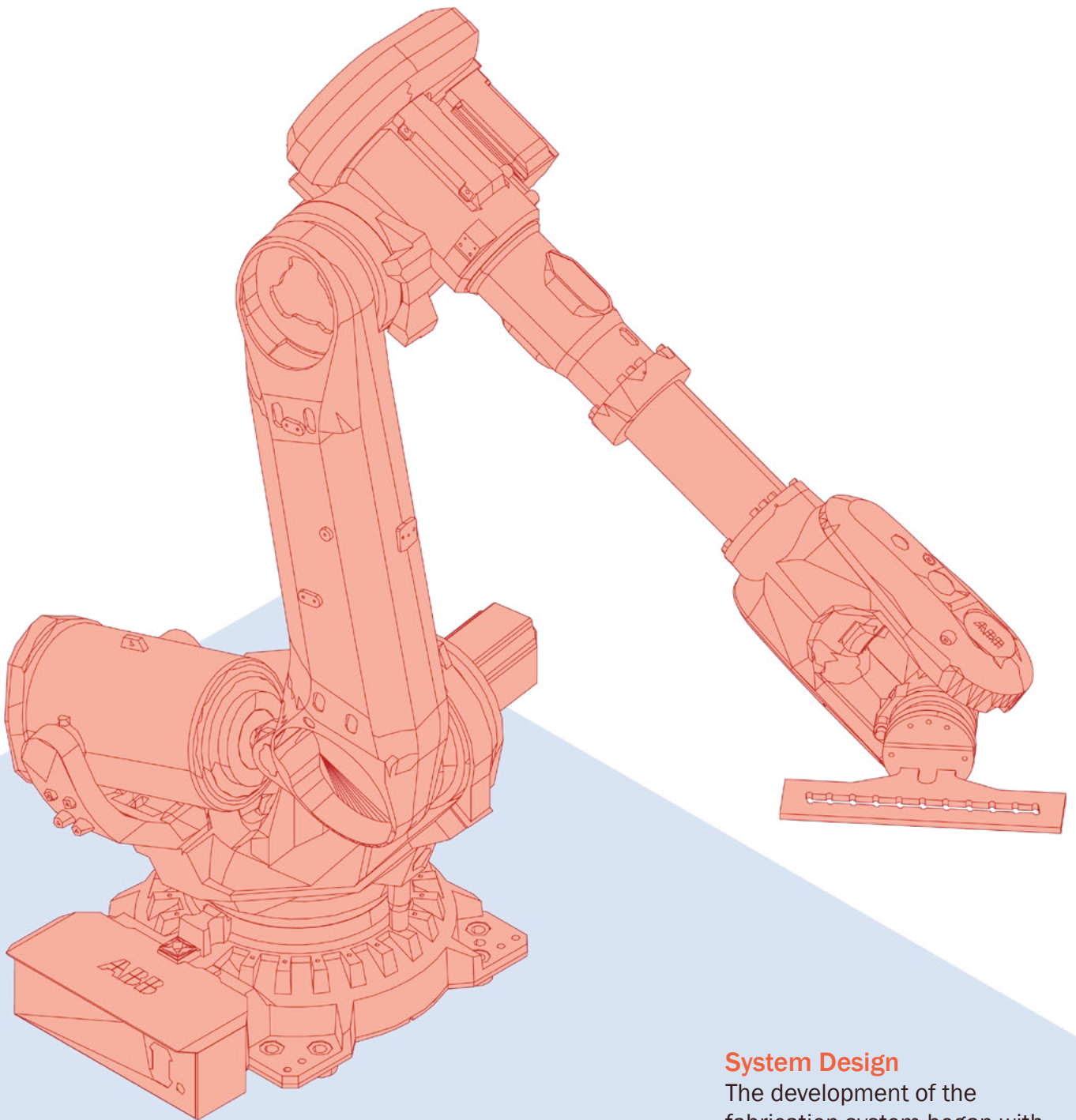


Figure 3.1 ABB IRB 6700 robot, axonometric diagram

System Design

The development of the fabrication system began with the design of the overall setup. The parts that were designed and fabricated were all specifically for the robot model ABB IRB 6700 (see Figure 3.1). Considering the facilities and equipment available during this thesis' timeline, the main structure pieces for the system were made with plywood that was milled with the CNC machine to achieve most precision.

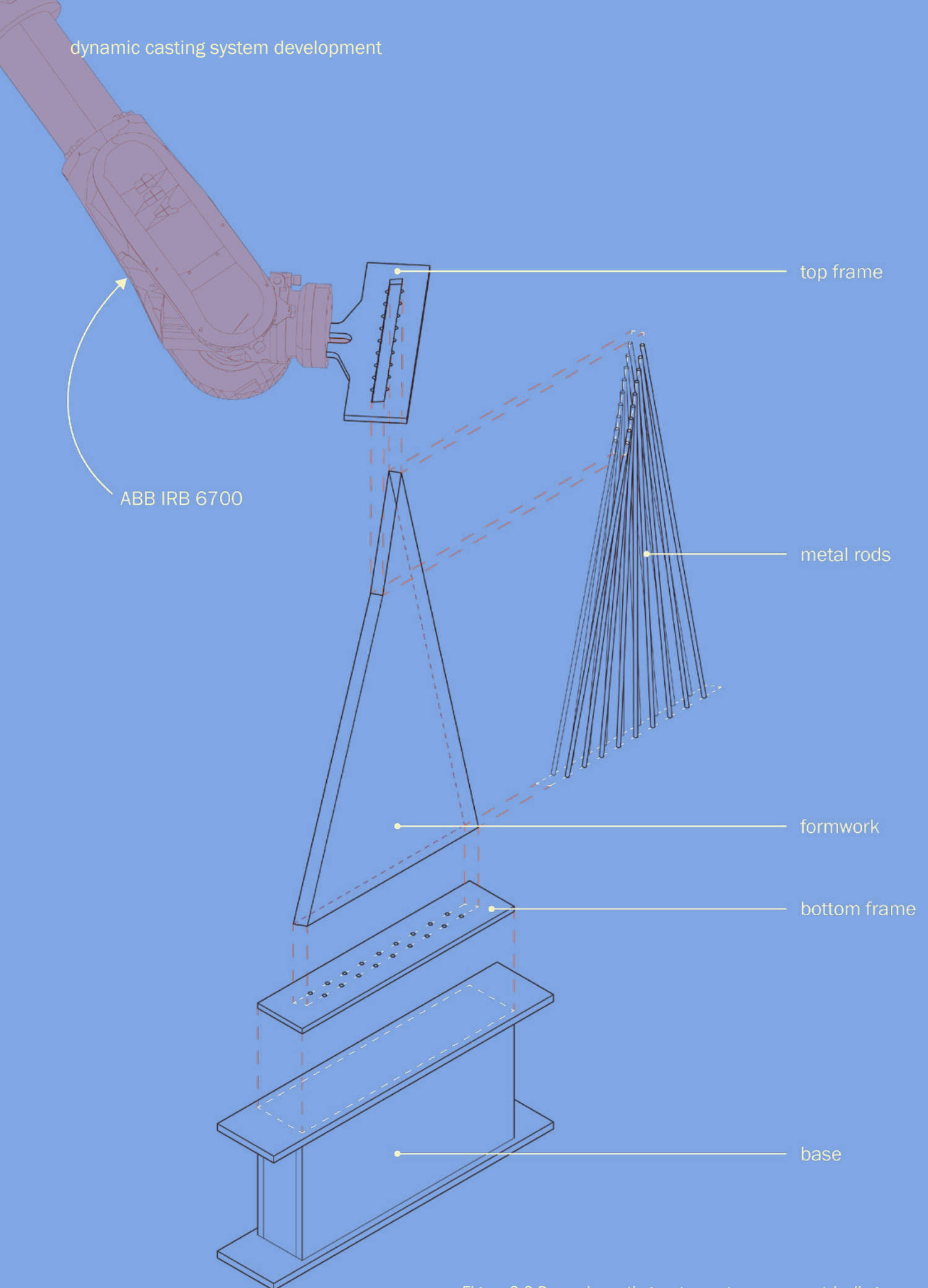


Figure 3.2 Dynamic casting system setup, axonometric diagram

System Parts

To simplify the system as much as possible, the overall setup is as follows (see Figure 3.2): There is a base, which is fixed to the ground for stability and elevates the formwork to a more accessible working height, facilitating casting and the subsequent removal of the component. On one hand there is a bottom frame attached to the base, which remains static during fabrication. On the other, the top frame is attached to the robot, acting as the end-effector, which is the moving piece that shapes the fabric formwork. The fabric formwork is attached to both the top and bottom frames securely with the steel rods slotted between them, acting as the ruling of the geometry. The rods are an integral part of the system because, not only do they counteract the hydrostatic pressure of the geopolymer as it's being cast into the fabric, but with the simultaneous movement of the robot (up in the z-axis while rotating), they help push the material into all the crevices and settle evenly. These are placed externally around the fabric formwork and are removed once the piece is hardened and cured.

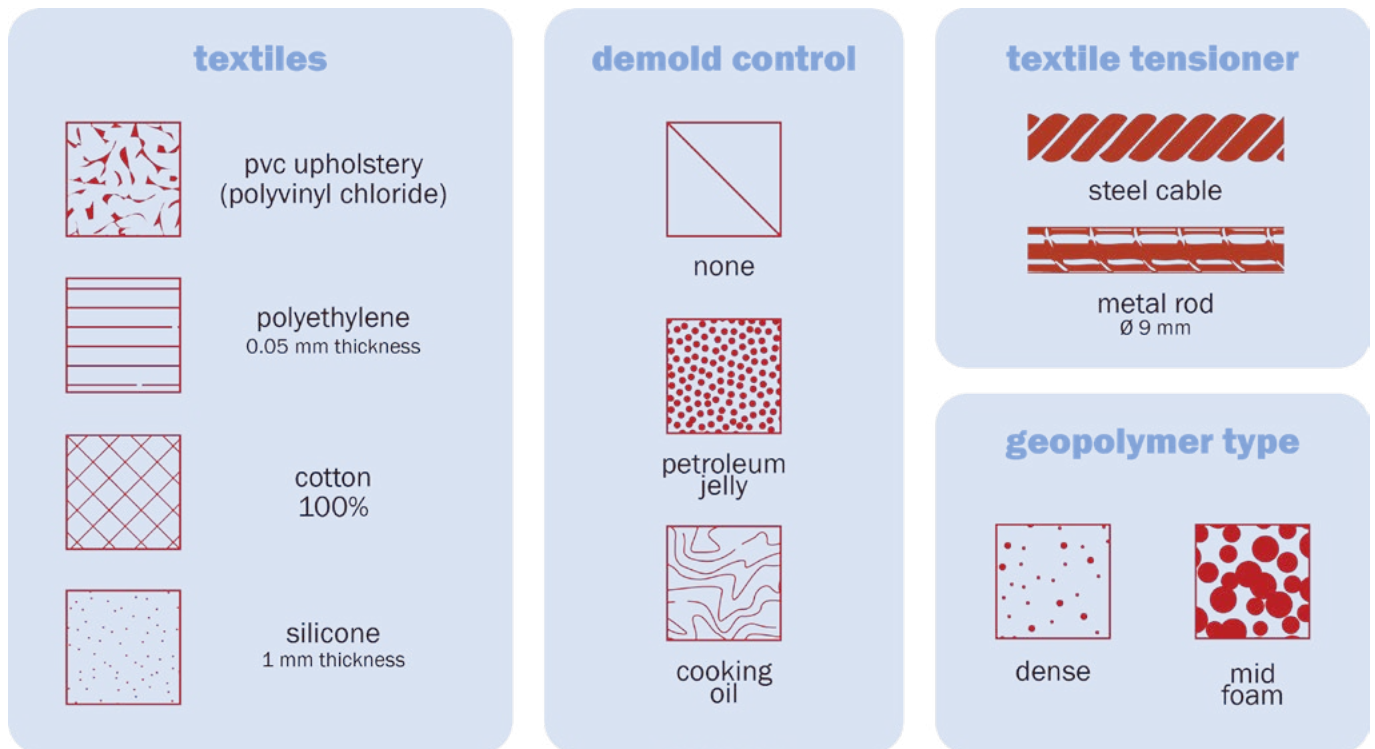


Figure 3.3 Variables and parameters evaluated for system design

Evaluated Parameters

Before arriving at that setup, the first step to designing the system setup was defining the materials that would be used to make each part. To do this, there were several variables that were tested against four main parameters in relation to the material behaviour. There were three main variables that were tested. Firstly, the textile used as the formwork material. Then considering the types of textiles selected, the second variable was a demould control; taking into account the curing process of a geopolymer, it was important to consider how to demould the piece once it was hardened. All the textiles were tested with both the dense and the mid foam geopolymer. Lastly, with the understanding of a geopolymer's viscosity when first made and poured, a textile tensioner was also tested. This was meant to help the fabric hold its shape after material is poured into it, to counteract the hydrostatic pressure of the uncured geopolymer.

There were four textile options that were tested: PVC upholstery (polyvinyl chloride), polyethylene, silicone and cotton (see Figure 3.3 for further details). The choice behind these options was mainly considering their flexibility while still being resistant. The second variable in consideration was a demoulding strategy. Given the geopolymer's reaction to adhere to most surfaces, it was crucial to also test different solutions on the only textile that did not exhibit low-surface-energy behaviour: cotton (Roosta, Zendeboudi and Rezaei, 2025). Unlike the other textile options, this material is wettable and porous meaning that liquids are likely to spread across their surfaces (Chaudhury, 1996); in the case of a geopolymer, the reaction of the gel structures occurs

while binding to the fibres of the cotton. Then, in order to maintain the shape of the hyperbolic paraboloid throughout the casting process, two types of tensioners were used: steel cables attached with screw hooks and steel rods that were slotted into holes.

The parameters that were evaluated in these sets of tests were the following:

First, the demouldability was tested. The main observation was to determine how easily the textile would peel back from the geopolymer after it had cured. The second parameter to test was the surface finish. Considering that the idea was to allow the textile to also be a design factor in the final outcome of the component, the textiles were tested to see what the surface texture would look like on the geopolymer. Third was to evaluate the textile conformability and observe how the textile adapted to forming conditions without creating wrinkles or folds. This was important because depending on the way the textile folded or stretched, the ridges it created could affect the process of demoulding the piece. The last parameter that was tested in the second set of tests was the textile tensioner. The main observation was to see how the surface control would resist the hydrostatic pressure of the geopolymer after being cast inside the formwork.

The remaining tests and prototypes were made using Sample C, which is UEM that was extracted from an ongoing construction site in the neighbourhood of Poblenou, Barcelona (see Figure 3.4).



Figure 3.4 Sample C extraction site, photograph



Figure 3.5 Experiment A setup, photograph

Testing Outcomes

For Experiment A, the parameters evaluated were demouldability, surface finish and conformability, as explained earlier. The methodology for this experiment started with cutting a square sample of each textile of about 25 x 25 cm. Then, the moulds for the previous experiments from Chapter 2 were reused as stands and holders. The textile square was pushed into the hole of the mould from the centre point to create a small cavity leaving the edges of the textile hanging from the sides of the square mould. This contraption was held by two other moulds to leave room underneath for the mix to settle and let the textile fully engage with the geopolymer. Finally, after securing the setup, the geopolymer mix (dense recipe) was poured inside and covered for curing (see Figure 3.5). After demoulding these the observations were recorded in the table on page 45.

Following the observations from Experiment A, the best performing textiles were tested with the same methodology but with the mid foam recipe geopolymer. The idea was to observe how the material would expand within this formwork configuration. Similar to the experimentation from Chapter 2, these were poured at about 75% of the full volume and all expanded similarly. Figure 3.6 shows the samples from this comparison.

Experiment A - Results Observations

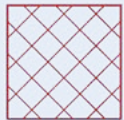
textile	demouldability	surface finish	conformability
PVC upholstery	Very easy	Shiny and smooth	The flexibility of the textile creates a lot of folds and wrinkles, but stretchable
polyethylene	Very easy	Glossy and smooth	The textile creates hard folds that become trapped once the cast material hardens
cotton	Hard; fabric adhered to the geopolymer	--	--
cotton + petroleum jelly	Easy	Knit texture imprint	The textile's softness and flexibility allows the cast material to mould and stretch into the material with minimal creasing
cotton + cooking oil	Easy	Knit texture imprint	The textile's softness and flexibility allows the cast material to mould and stretch into the material with minimal creasing
silicone	Very easy	Shiny and smooth	The stiffness of the textile keeps the form sturdy, but creates hard creases when twisted; not stretchable



pvc upholstery



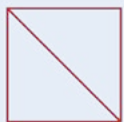
polyethylene



cotton



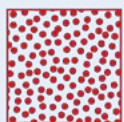
silicone



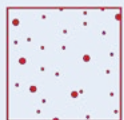
none



cooking oil



petroleum jelly



dense
geopolymer



mid foam
geopolymer



steel cable



metal rod

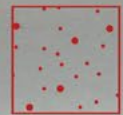




Figure 3.6 Comparison of dense versus mid foam geopolymer in Experiment A

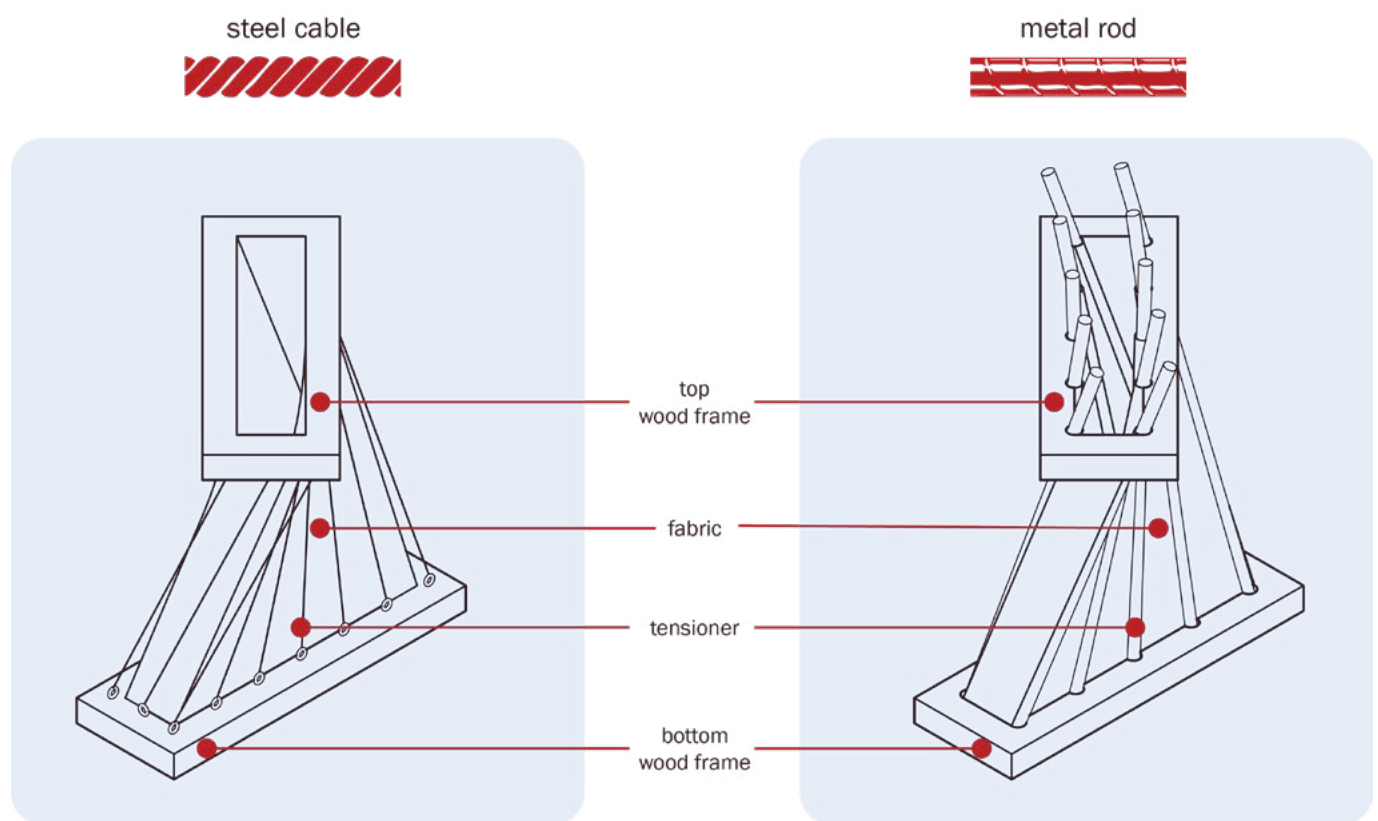


Figure 3.7 Experiment B setup, diagram

In Experiment B, the main parameter evaluated was the textile tensioner. Before scaling the fabrication system up and setting up the robot, it was important to ensure that the chosen materials for each part of the system were suitable. The methodology for this one was to simulate the final rotation that the robot would make, so 2 frames were cut from plywood, with an inner dimension of 320 cm and an edge of 2 cm. A piece of the chosen textile was cut with a length of the frame's inner perimeter and a height of 25 cm. The tensioners were added and then the top frame was rotated to the desired angle. The frames were held in their rotated position by a vertical plywood element secured between the top and bottom frames with screws.

As a result, three variations were made at a small scale to observe how the geopolymer would settle inside the formwork with the tensioners. Each was made after observing the previous one's performance. Using the steel cables was the initial instinct. B1 used PVC upholstery as formwork and the steel cables with the screw hooks. The main observations from this one were that the steel was too stiff to knot around the hooks with tension, which made the formwork stretch a lot, leading to a component a lot thicker than intended. This led to B2, which

changed the textile to silicone, which is thicker and sturdier. The prediction was that this would help stabilise geometry better using the steel cables. However, this was not the case; similarly to B1, the cables were not tight enough, which led to the slumping of the material to the bottom of the piece. B3 produced the most suitable result and was consequently selected for further development. After realizing that the cables were not sturdy enough to contain the material inside the formwork, the task was to find a rigid yet thin material that would shape the component. As opposed to the steel cables, the metal rods were already stiff and rigid in a straight line without needing additional tensioning. These were used by slotting them into frames with the freedom to move and rotate with the movement of the formwork. Both B1 and B2 created a variation of ridges and folds after twisting that were very difficult to control. For that reason, B3 used cotton as the textile material. This configuration performed successfully because the cotton stretched sufficiently to accommodate the material while the steel rods controlled excessive deformation and maintained a relatively even thickness. The rods also created a ribbed texture that drew the ruling of the hyperbolic paraboloid, further emphasizing the geometry. See Figure 3.8 for the visuals of these results.



steel cable



metal rod

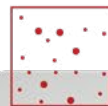
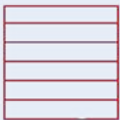


Figure 3.8 Experiment B results



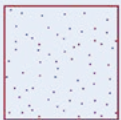
pvc upholstery



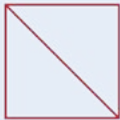
polyethylene



cotton



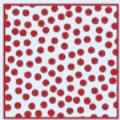
silicone



none



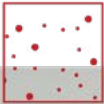
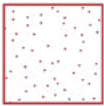
cooking oil



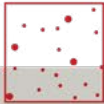
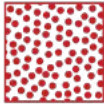
petroleum jelly



dense geopolymer



B2



B3

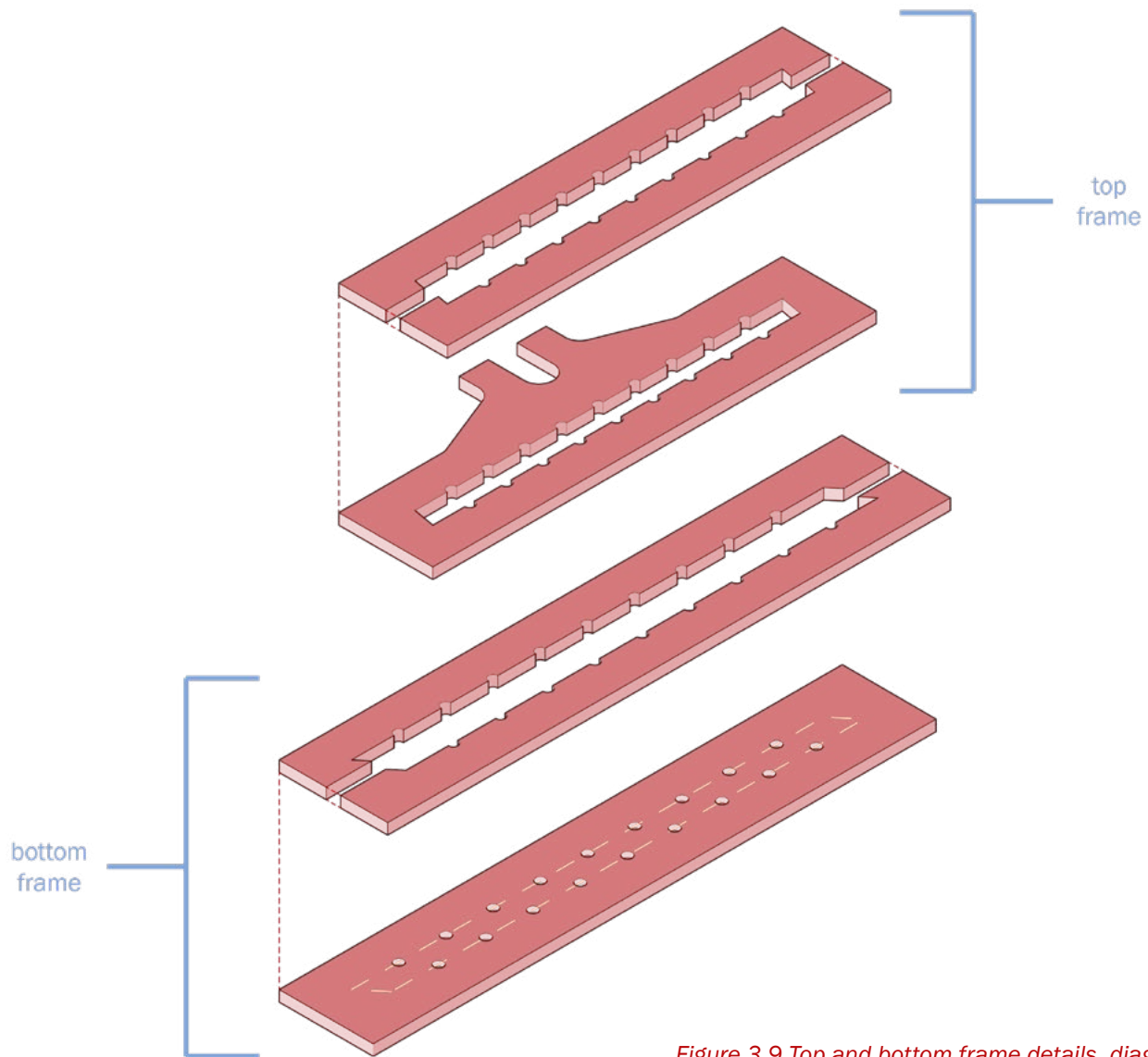


Figure 3.9 Top and bottom frame details, diagram

Prototype Testing

After finalizing the materials and mechanism to use in the fabrication system, the next step in the research process was to scale up (see Figure 3.2 on page 40 for reference). As previously described, the material used for the frame system was plywood, the formwork was cotton and the tensioners were the steel rods.

The mechanism used in the frames is slightly more nuanced than in the diagram shown in Figure 3.2. Each frame comprises two matching plywood layers that clamp

the upper and lower edges of the fabric. The purpose of this second piece (for both top and bottom) is to seal the fabric using the extra flaps of fabric that were intentionally designed into the formwork, which are pressed between the two pieces of the frame (see Figure 3.9). The pieces of each frame were fastened together using screws, which kept the fabric clamped tightly in place; this ensured no material would leak through the gaps and also allowed the geometry to stay precise after the robotic movement. Moreover, the bottom-most frame piece (the

first piece from bottom-up in Figure 3.9), has some holes that were milled with the CNC during the fabrication process. These are slightly larger in diameter than the metal rods to slot them in place and allow them to move as the top frame moves and rotates. The remaining pieces from the frames all have grooves that are meant to keep the rods in place during the fabrication process.

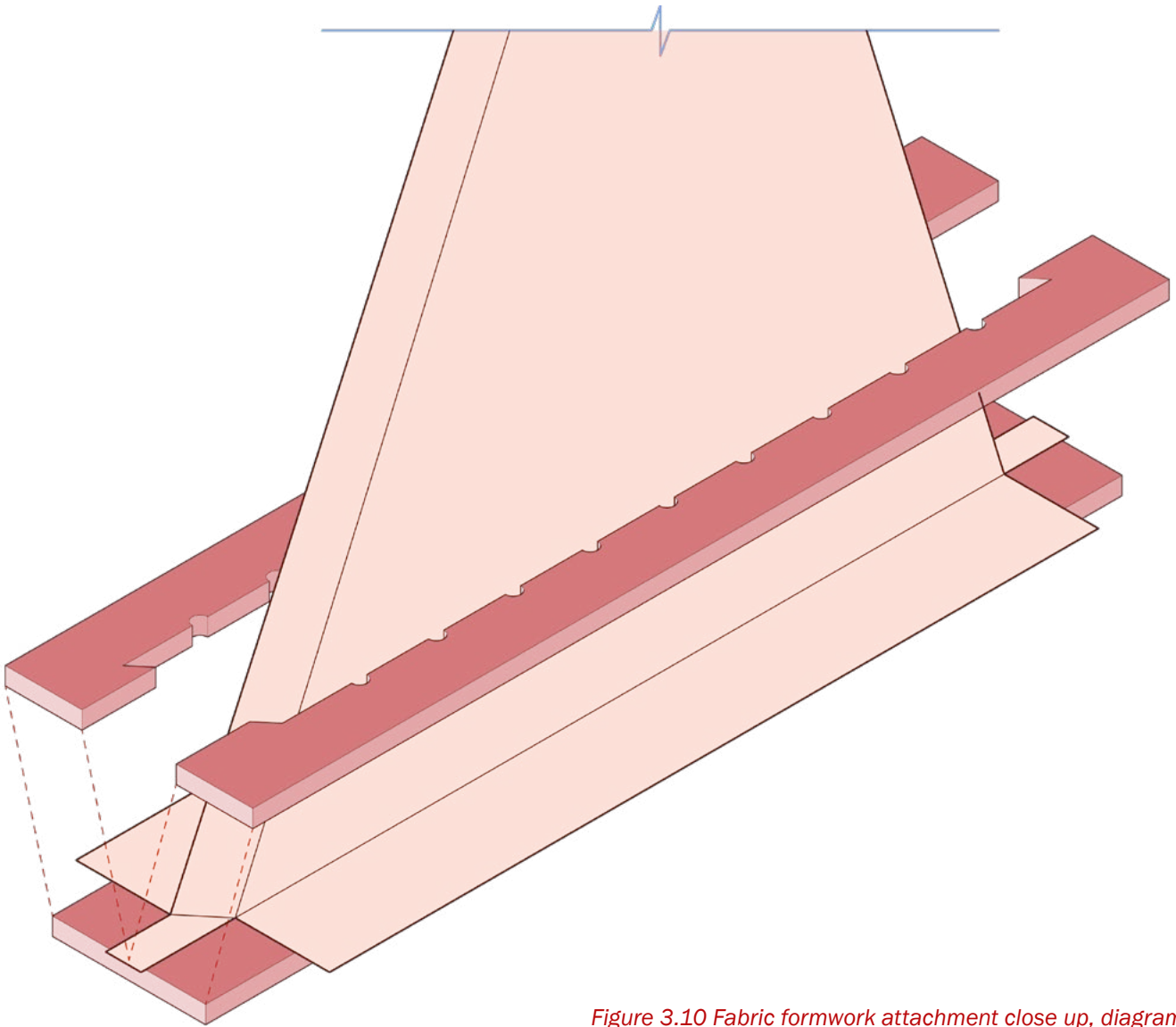


Figure 3.10 Fabric formwork attachment close up, diagram

Figure 3.11 Unrolled pieces of fabric formwork, diagram

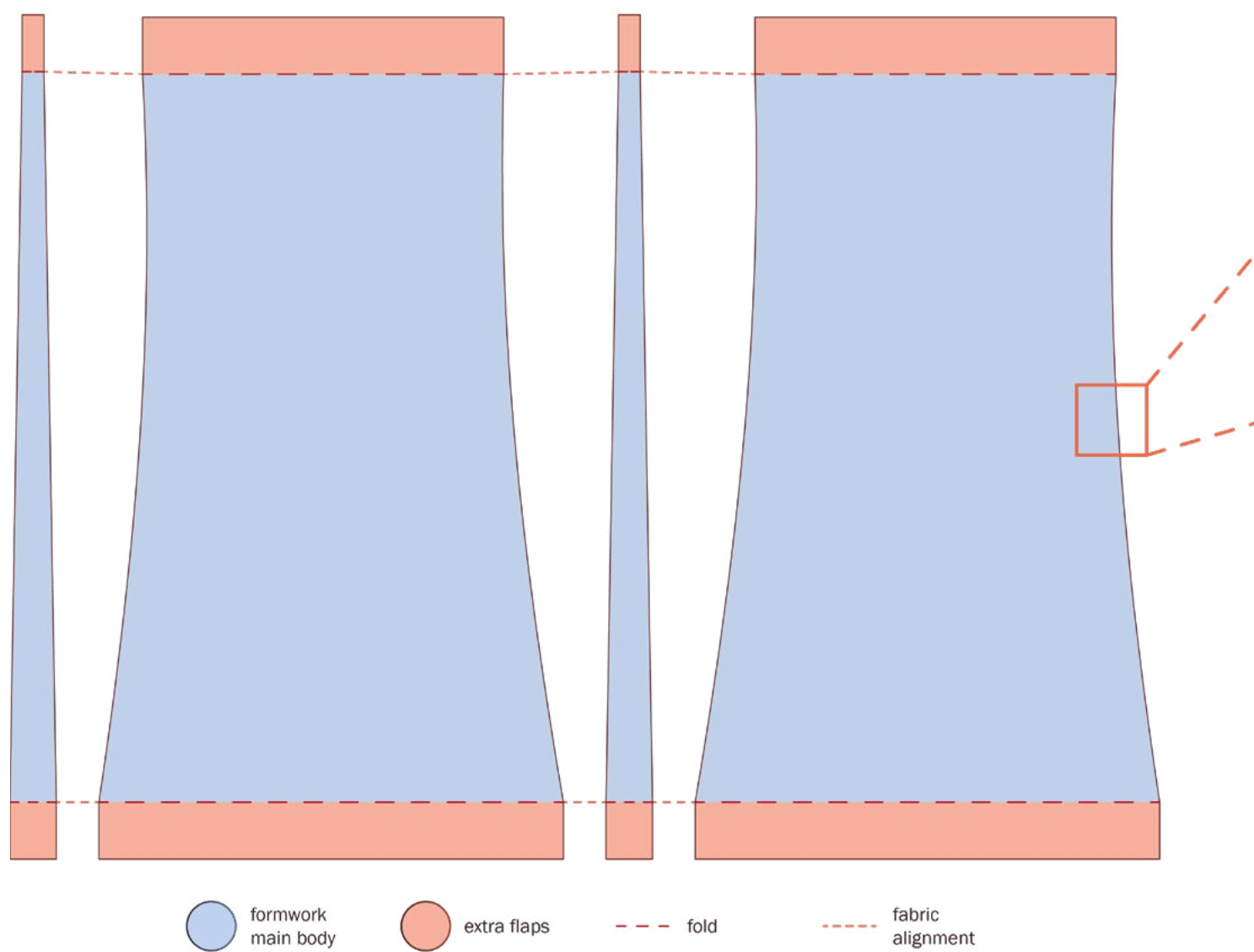




Figure 3.12 Sewing a fabric formwork prototype, photograph

To ensure the vertical alignment of the hyperbolic paraboloids once cast and demoulded, the top and bottom frames were designed for a specific rotation, which is why the length differs. Taking this into account, the fabric formwork was designed accordingly (see Figure 3.11). The slight curvature of the fabric pieces was derived from a digital simulation of the final hyper geometry. There are 4 main pieces of fabric that are cut and sewn together (see Figure 3.12). Another reason for choosing cotton as the formwork textile was to enable use of waste fabric. As a result, for the prototypes developed, old cotton t-shirts were cut and sewn to make the fabric formwork.



Figure 3.13 Test prototypes, photographs

Learning Outcomes

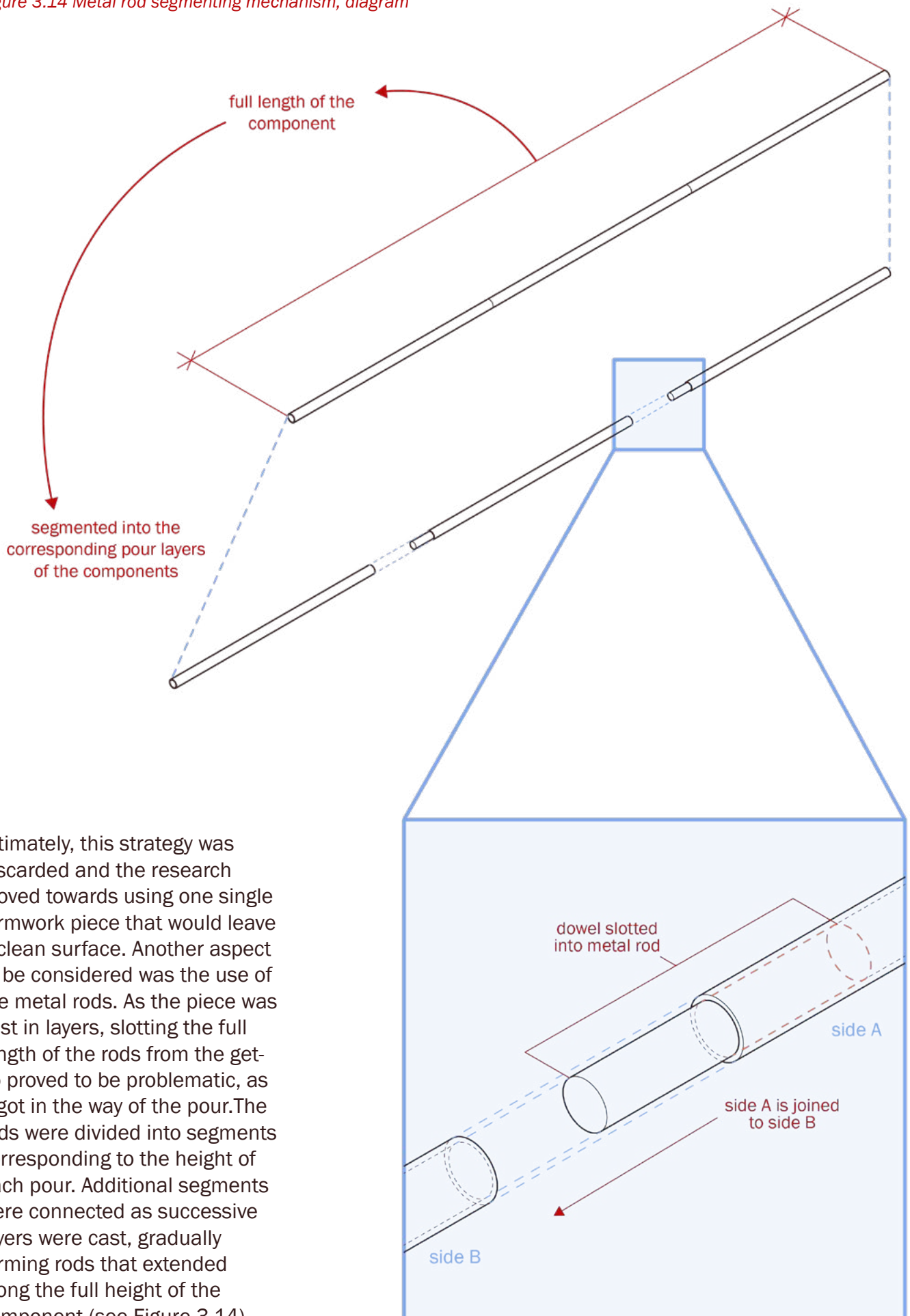
Frame Dimensions

The purpose of implementing the double curvature and ruled surface geometry was to optimise the structure and the material. One of the main goals, from the beginning, was to use the least amount of material; in other words, the idea was to make each hypar component as thin as possible. In the process of prototyping test components, some were too thin and consequently fragile, while others were excessively thick and heavy. Finding a balance between them was the objective, while taking advantage of the material gradation developed to also help lighten the component even more. Another factor to consider was the viscosity of the geopolymer and the fact that there were different sized particles. A frame with a very thin opening would not allow the material to properly settle inside the formwork.

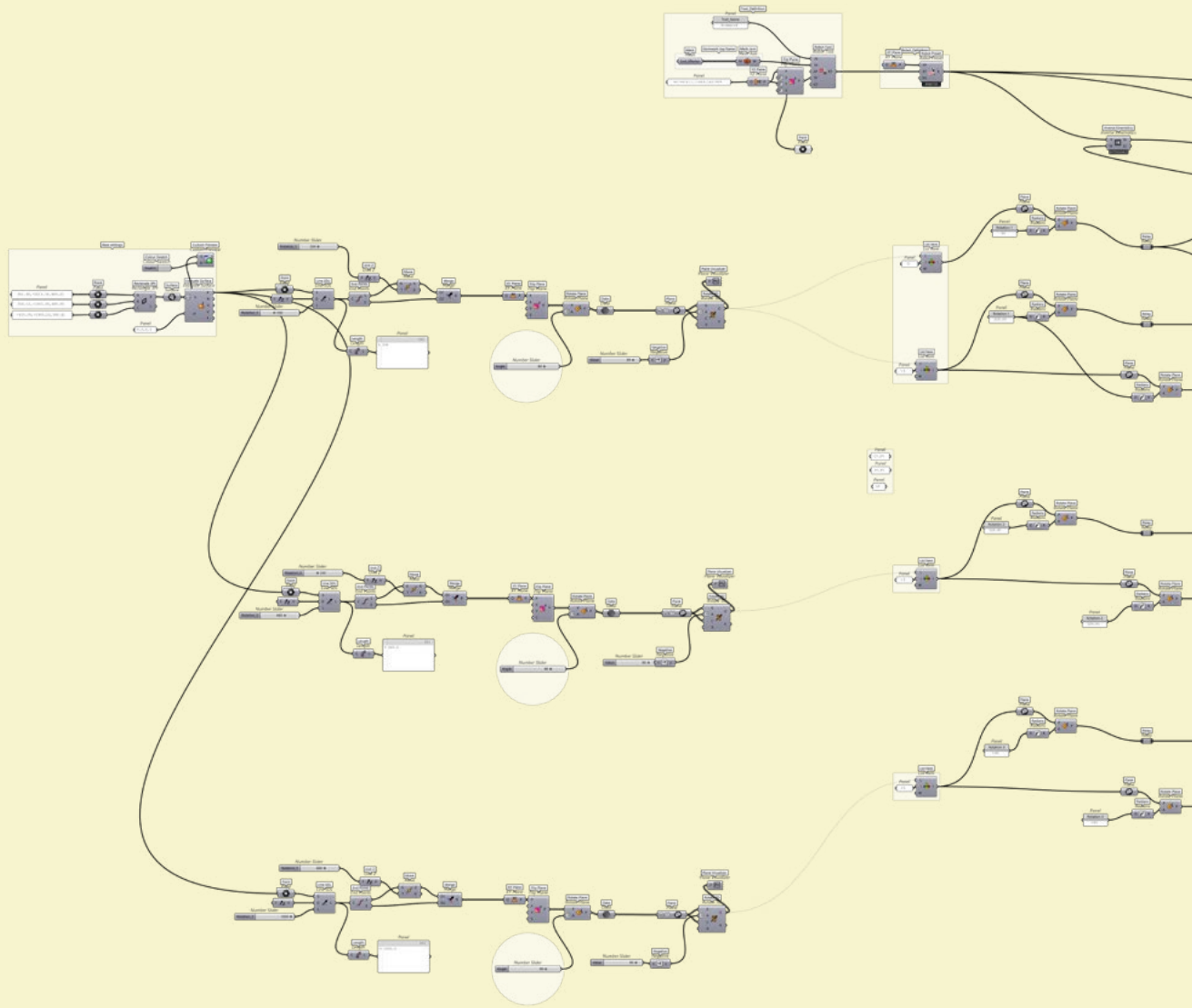
Pouring Strategy

Another step that was tested differently was the pouring of the geopolymer. It was already established that in order to develop the functional gradient, the geopolymer would be cast in layers accordingly. One of the initial strategies was to divide the whole hypar into parts with their corresponding fabric formwork and then stack them vertically. This proved to be very problematic because the alignment needed to be incredibly precise and the lack of continuity created a non-uniform surface, also susceptible to cracking.

Figure 3.14 Metal rod segmenting mechanism, diagram



Ultimately, this strategy was discarded and the research moved towards using one single formwork piece that would leave a clean surface. Another aspect to be considered was the use of the metal rods. As the piece was cast in layers, slotting the full length of the rods from the get-go proved to be problematic, as it got in the way of the pour. The rods were divided into segments corresponding to the height of each pour. Additional segments were connected as successive layers were cast, gradually forming rods that extended along the full height of the component (see Figure 3.14).



Scaling Up

One of the challenges that came up during the scale up process was the hydrostatic pressure of the material. The component is cast continuously, without letting each layer cure, to promote interlayer bonding. As a result, as each layer is poured, the hydrostatic pressure increases and the bulging and slumping of the fabric formwork also increases. For a larger piece, the number of steel rods was increased to better contain the shape. Moreover, to secure the rods in place, additional fabric flaps were made with holes in them to pass the rods through so they would not slide off while the robot moved.

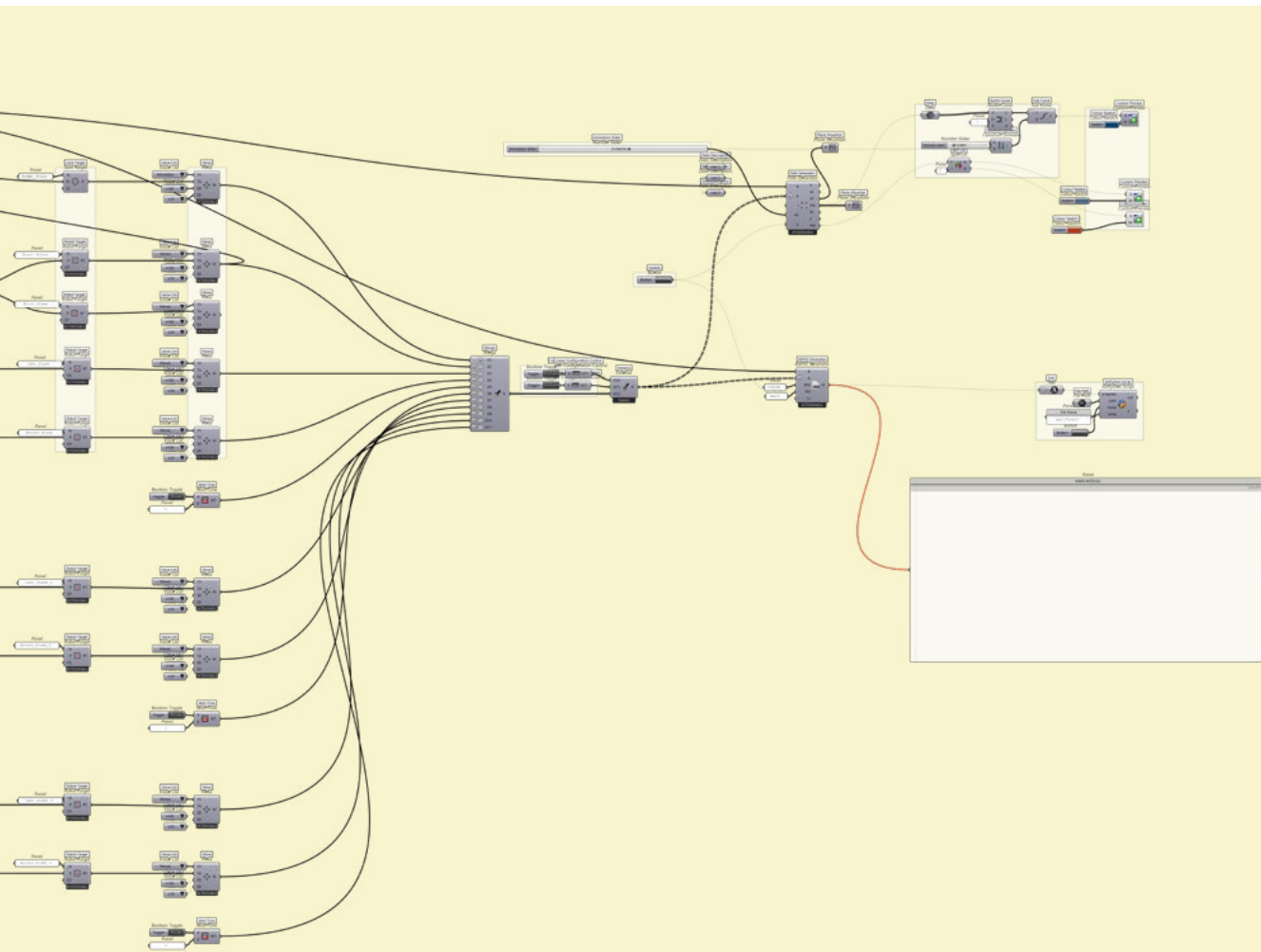


Figure 3.15 Grasshopper script overview, screenshot

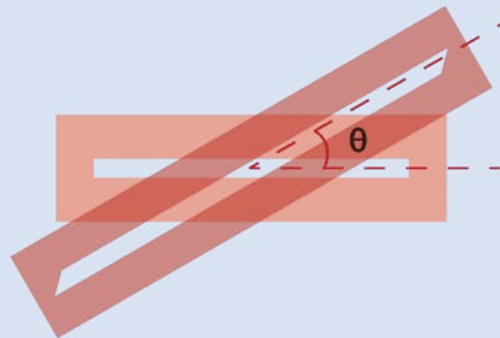
Robotic Script

The calibration of the robotic script was an integral part of the entire system development. To automate the robotic movement, a script was developed using Grasshopper with the Robotic Components ABB plugin. Not only does it enable the creation of the code to feed the robot, but it also creates a digital simulation of the whole movement, making it easier to make adjustments. The robotic programming also underwent various iterations. Initially, the movement was scripted such that the end-effector would move up in the z-axis, pause and then do the first rotation, before repeating the process until it reached the full height and rotation angle. Although this worked, it was noted that it slowed down the process given that the speed of the robot had to be maintained at a lower value for safety reasons as the metal rods were slotted in place and stacked up. The later iterations of the script were further optimised such that the end-effector moves up in the z-axis and rotates simultaneously until reaching its final position.

At this point of the research, it was important to acknowledge the key role of the robot in this system, and ask: **why use robotic fabrication to do this?** There are three main elements that answer this question.

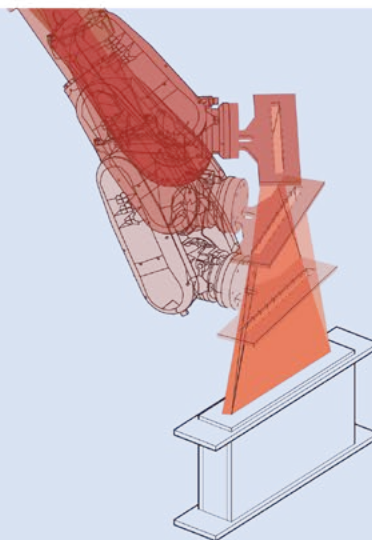
Firstly, there is the element of precision. As the ultimate goal is to create a structure made up of components that are assembled together, the alignment of these is crucial. The overall design can be completed digitally, and with tools in Grasshopper, these can then be segmented into components that are fabricable. Whether the design has a variation of hyperbolic paraboloids or they are repeated, the angles of each component have to be done with precision. Programming the movement digitally and transferring it to the robot ensures that this is achieved.

Secondly, robotic movement enables continuous geometric transformation. The fundamental aspect of the system is the controlled movement of the formwork, which not only shapes the



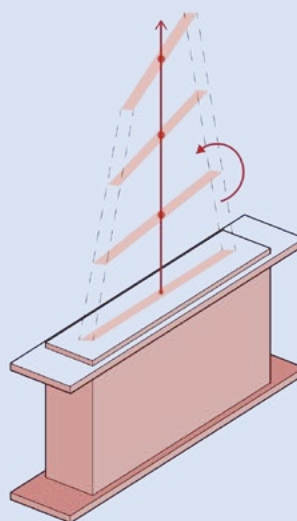
angle/rotation precision

precision



dynamic casting

dynamism

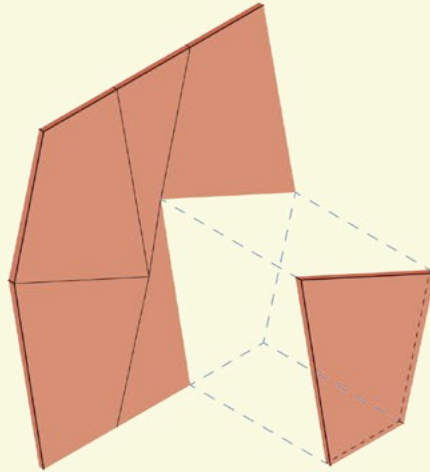


programmed trajectory

efficiency

precision

component alignment

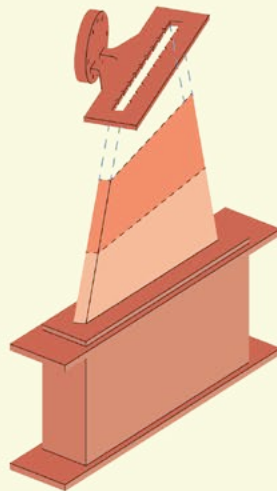


component but also helps distribute the material within the textile enclosure. The casting process can be done in stages resulting in the functionally graded finished component. This is possible because of the pause included in the script after the simultaneous translation and rotation movement.

Lastly, the use of robotic fabrication creates efficiency. Establishing a programmed trajectory for each component creates the potential for a repetitive production workflow. Considering that the goal is to achieve architectural scale structures, the robot also supports and maintains the position of the upper frame during casting, reducing the manual effort required to control larger formwork assemblies. Once the movement is complete, the robot can maintain the position of the upper frame while temporary vertical supports are secured between the frames. The assembly can then be detached from the robot and left to cure while the next component is prepared.

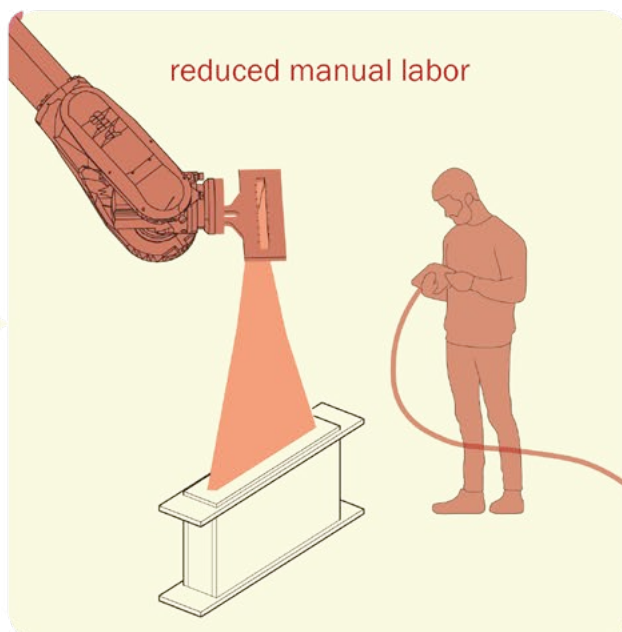
dynamism

casting in different stages



efficiency

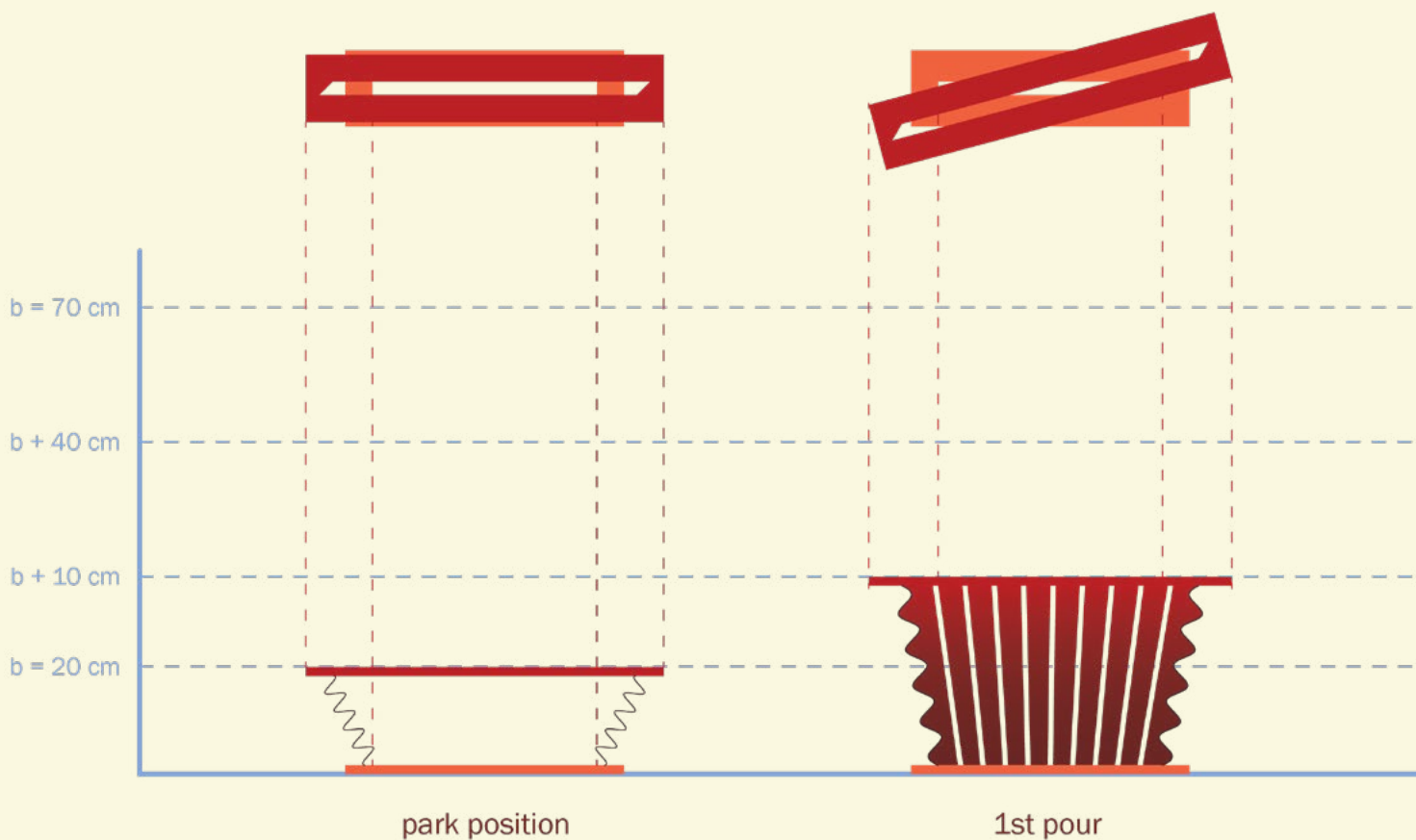
reduced manual labor

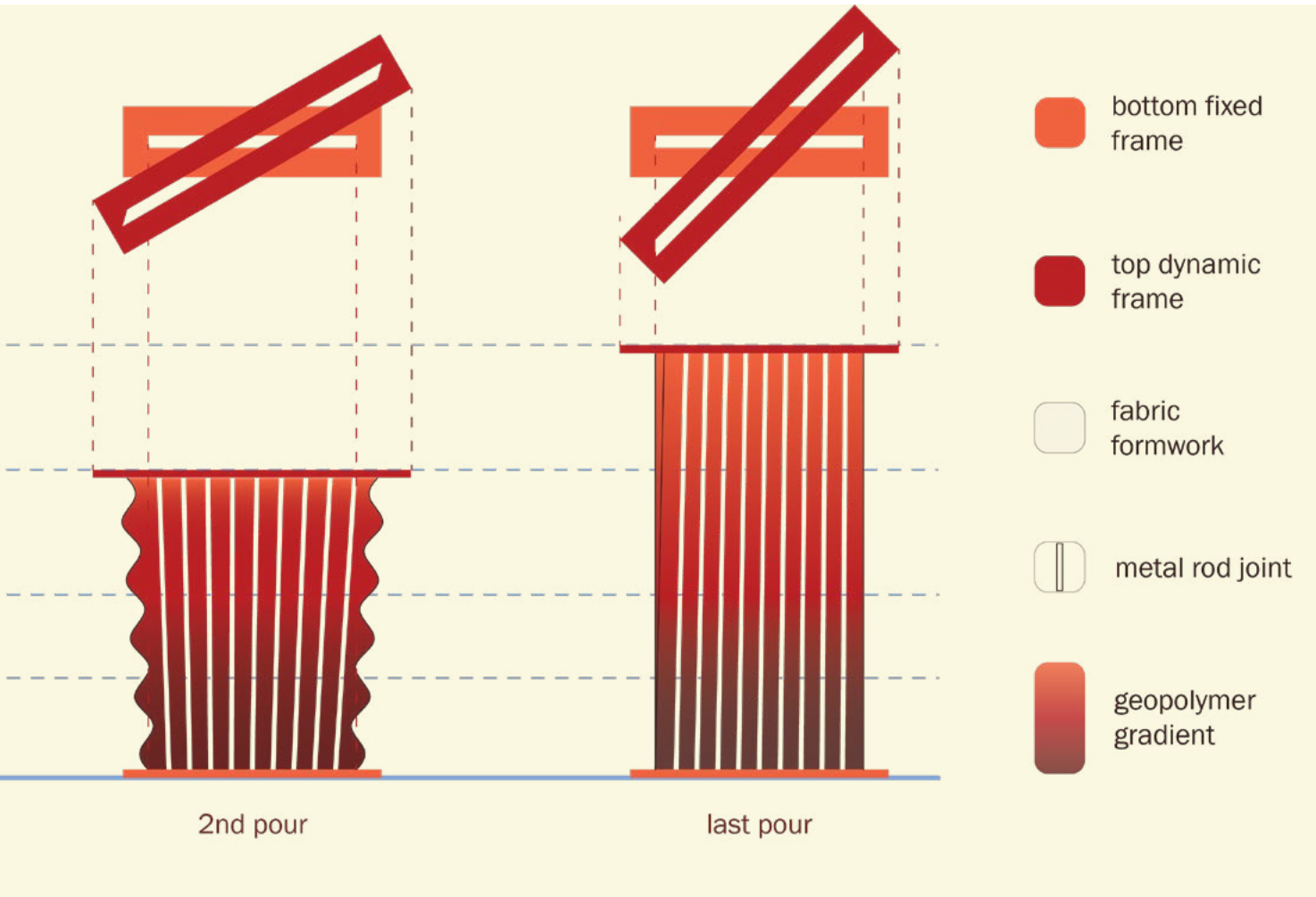


Finalised System

With the completion of the tests and learning outcomes, the fabrication system was finalized. It uses a singular formwork piece made of reclaimed cotton fabric. The robot is then programmed such that the top frame moves up in the z-axis and rotates simultaneously in stages to allow for the geopolymer layers to be cast until reaching the final position and shape. Figure 3.16 shows a diagrammatic representation of this process to illustrate the system.

Figure 3.16 Robotic movement and pouring stages, diagram (continued on next page)





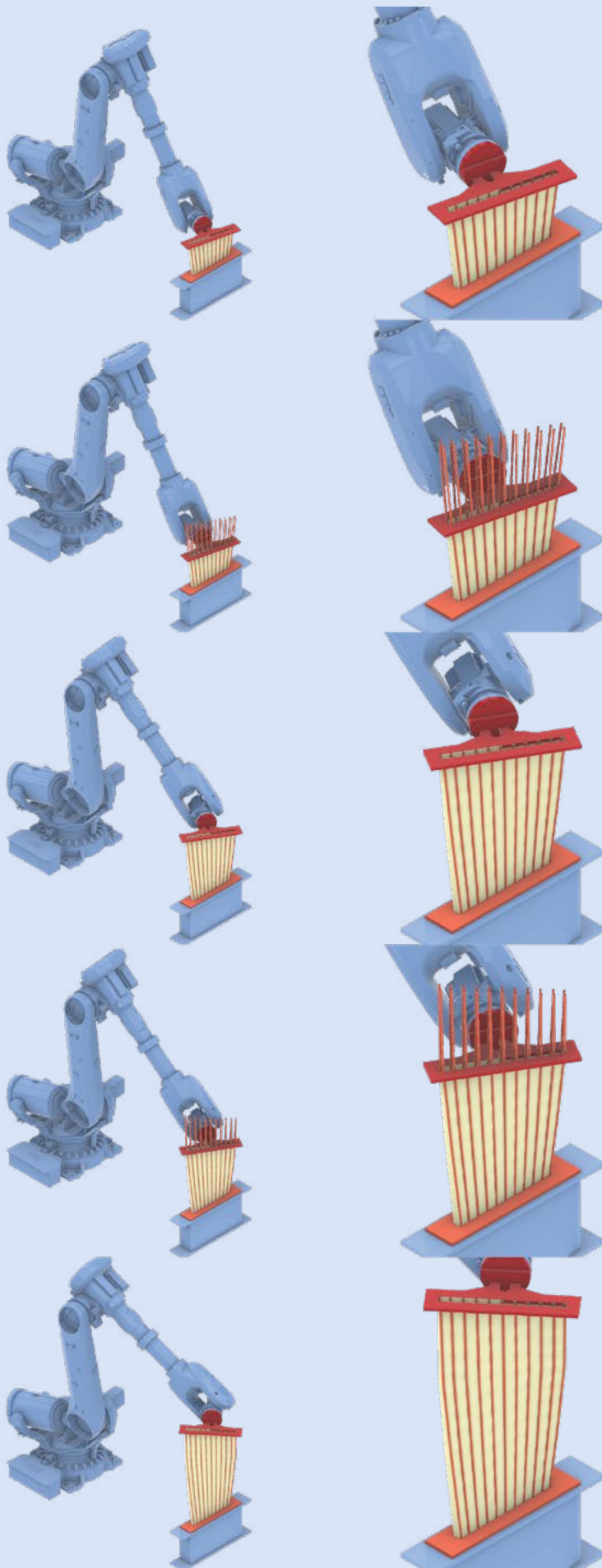


Figure 3.17 Robotic movement simulation sequence, Rhino screenshot

As shown earlier in this chapter, the bottom frame is attached to a base that elevates the entire piece, creating an offset to the ground as a safety measure to maintain the robot at a safe distance from collisions. The starting position of the top frame is also offset from the bottom frame to allow the metal rods to be slotted in and also because the fabric formwork is naturally in the way. Refer to Figure 3.17 to see the movement sequence from the digital simulation that was run before transferring to the actual robot.

As part of the research process and this thesis, a 1:1 scale prototype was developed as proof of concept. This scale refers to a full-size panel forming part of an architectural-scale assembly. With time and resource constraints, one hyperbolic paraboloid component was developed (see Figure 3.20). This piece had a width of 0.5 metres, a height of 1.0 metre and a thickness of 4 centimetres, measured between the rods disregarding the slight stretching of the fabric. It was composed of 3 geopolymer layers, meaning it was made with 3 pours consisting of the dense recipe with 10% of added metakaolin, followed by low foam and mid foam with the same base. After casting the piece, the top was sealed with another piece of plywood with petroleum jelly for ease of removal and to keep a flat surface at the top.

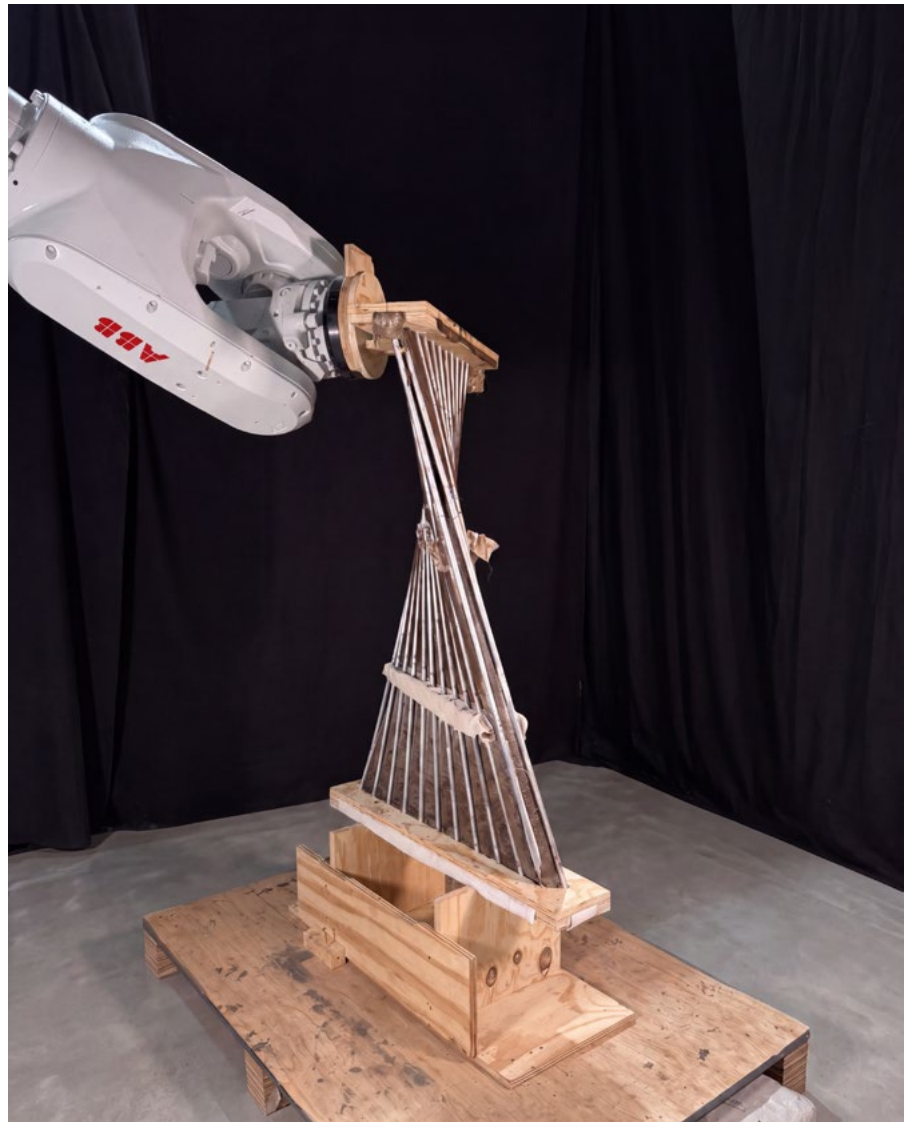


Figure 3.18 Final prototype prior to demoulding side view, photograph



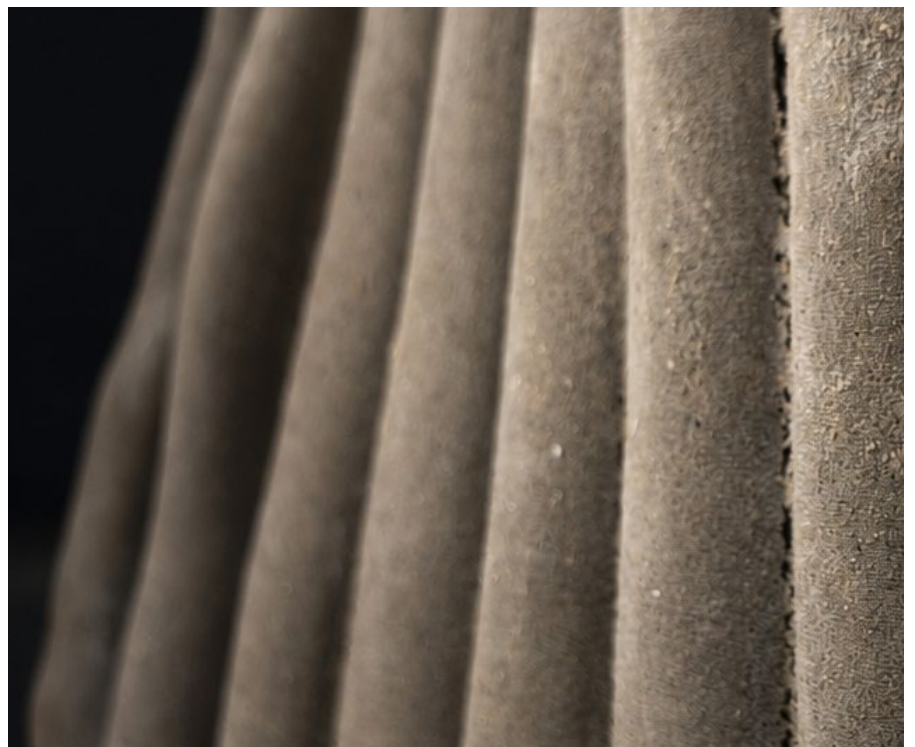
Figure 3.19 Final prototype prior to demoulding top view, photograph

Figure 3.20 Final prototype, photograph layout (across pages 66-67)



The images featuring this prototype (Figure 3.20) show a visible unevenness at the top of the piece; this occurred during the demoulding process due to a short curing period. Additionally, given that the geopolymer layer at the topmost section of the hypar was the foamed geopolymer, the porosity made it more susceptible to cracking without proper curing.





04

design application

Although this thesis has research spanning material and fabrication investigations, architectural application has served as the guiding principle throughout. The following chapter compiles all the research and findings from the preceding chapters, bringing it to life in the architectural landscape. It starts by outlining the connection designed into the panels that make it a component-based installation system. Then, a case study is presented with an array of visuals at multiple scales to better illustrate the potential that this system offers.

Assembly System

Earlier in the booklet, it was mentioned that the goal for this fabrication system is to create these hyperbolic paraboloid panels using the UEM geopolymer and assemble them into larger structures. With a prefabricated system in mind, the logic for the assembly is that there will be an overall design of the structure that is then segmented into smaller components that can be fabricated with the dynamic casting system and then transported to be assembled on site. With modelling tools such as Rhinoceros and Grasshopper, the design can be done and segmented digitally. Each segment is analysed as an individual component, after which its corresponding frames and fabric formwork are fabricated. In repetitive designs, identical components could be produced using the same frame and formwork configuration.

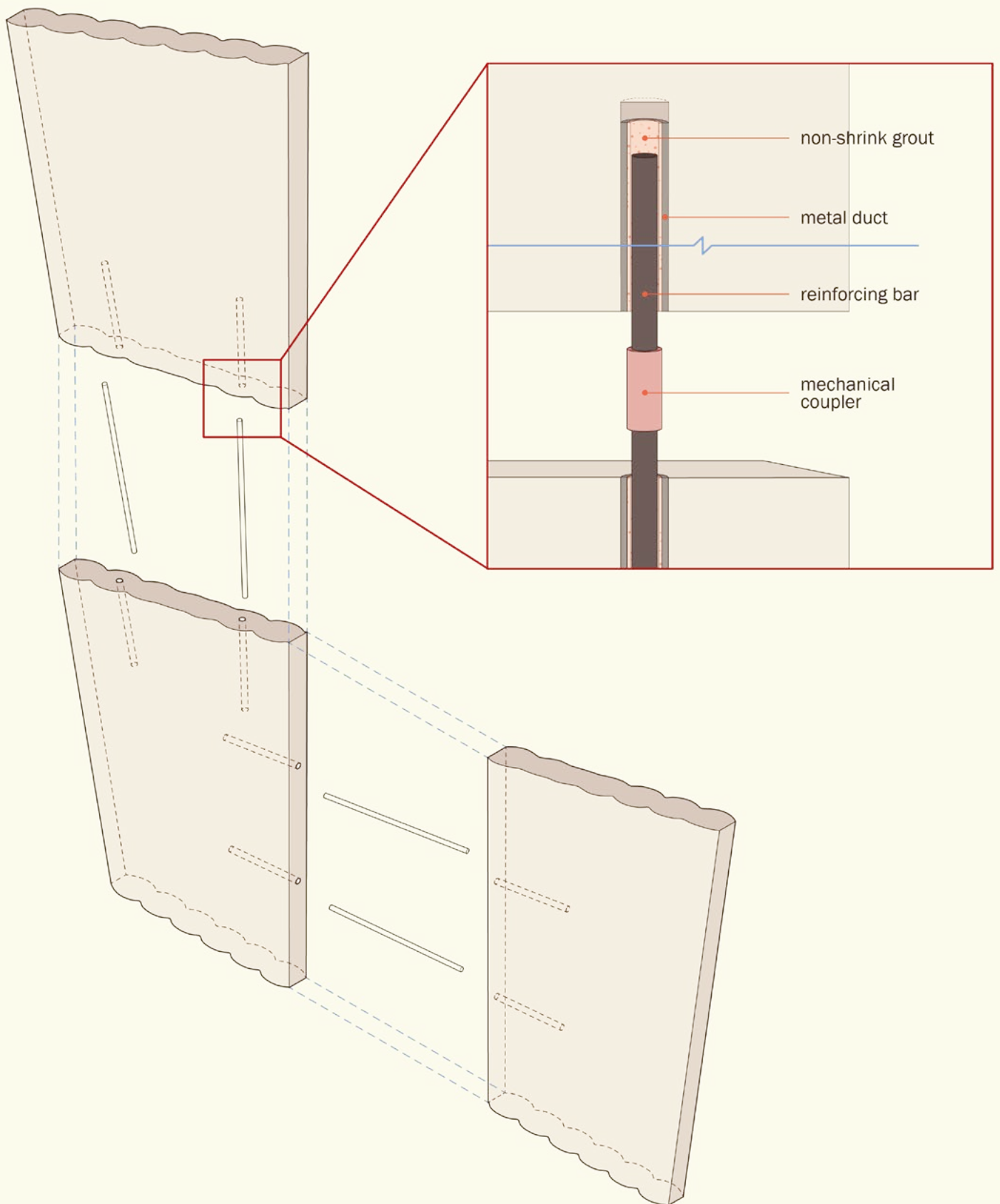


Figure 4.1 Panel-to-panel connection for UEM geopolymer panels, diagram

Component Installation

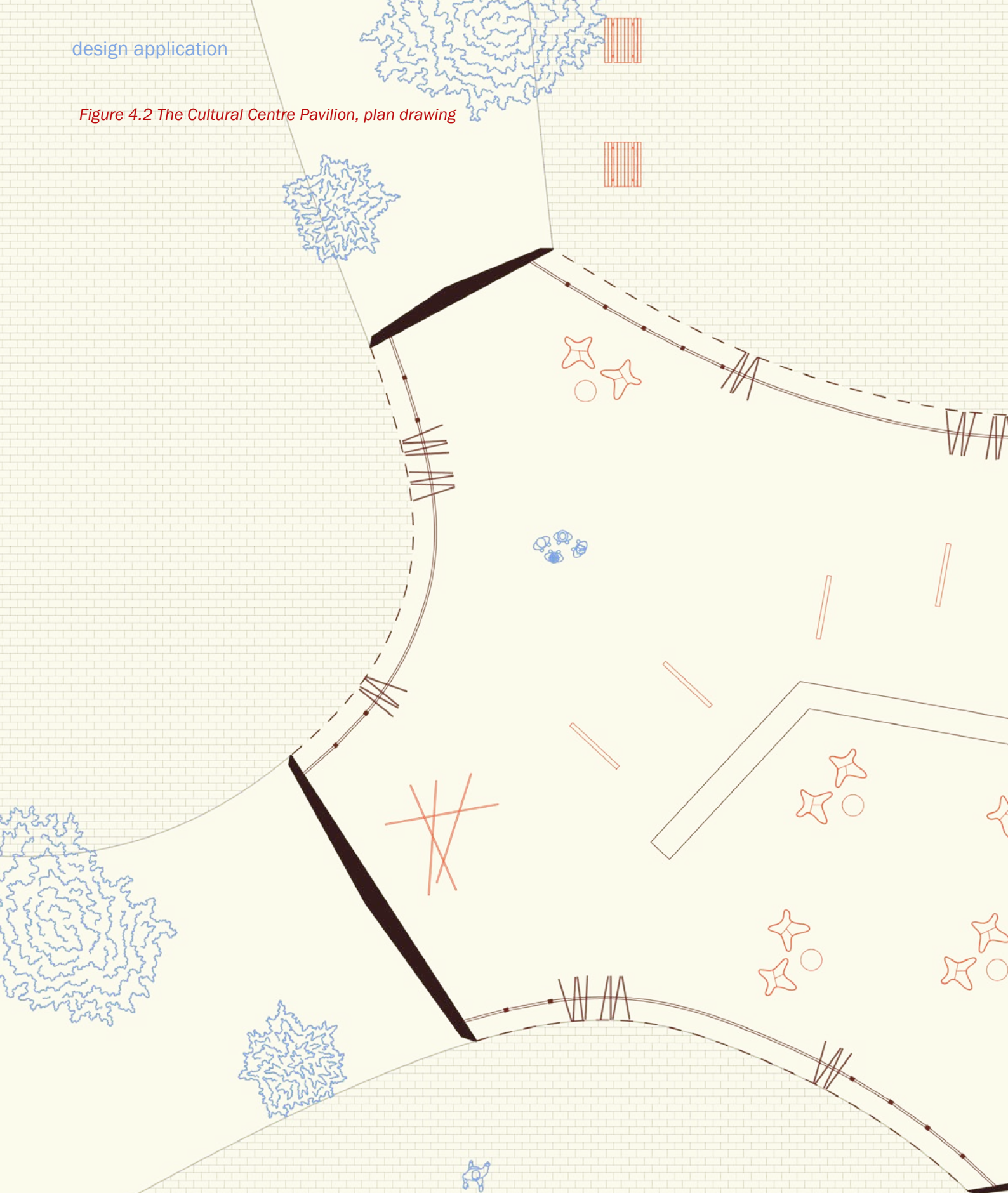
As previously established, the material behaviour of geopolymers, and more specifically in this case, of UEM geopolymers is very similar to that of conventional concrete: not only in its strength capabilities but also, in that it is a dense mix that hardens to a solid matrix. As such, the logic implemented in the design of the connection system of the cast components was derived from existing strategies used in prefabricated concrete panels. It is important to note that this remains a conceptual proposal that would require further testing and investigation.

The connection type proposed for the dynamically cast panels is an adaptation of a conventional grouted dowel-in-duct precast connection (Elsayed and Nehdi, 2017). Given the assembly logic for the component-based design used with the UEM geopolymer panels, this strategy can also be applied as they are cast as individual elements that are to be assembled. The principle is adapted to the prefabricated components by incorporating receiving ducts within designated connection zones. Elsayed and Nehdi (2017, p. 1), explain that “a large diameter reinforcing bar [...] is typically bridged over the horizontal joint between two vertical panels via a mechanical coupler”; “the extended length is then grouted inside a corrugated metallic conduit, which is embedded in the precast panel”.

Case Study: Cultural Centre Pavilion Introduction

The research culminated in the development of a case study that integrates the material system that was developed over the course of the investigation. This section of Chapter 4 introduces and details the Cultural Centre Pavilion, a digitally designed case study based on the dynamically cast UEM geopolymer panel system. The Cultural Centre Pavilion is an exterior public space that serves as an exhibition area, through which the fabrication system is explored at the scale of a public architectural setting. The pavilion was selected as the case-study typology because it allows the fabrication system to be explored beyond the scale of the individual prototype while keeping its material and geometric logic clearly legible. Its shell configuration demonstrates how multiple hypar modules can be aggregated to form a continuous structure in which the conventional distinction between wall and roof begins to dissolve. The structure came to fruition as the geometry of the panel itself informed the design of the shell. The pavilion’s open and publicly accessible character exposes the surfaces produced by the textile formwork, allowing their ribbed texture and material variation to become part of the architectural expression. The case study therefore operates not as a fully engineered construction proposal, but as an architectural demonstration of how modular aggregation, material grading and dynamic casting could collectively inform a larger spatial system.

Figure 4.2 The Cultural Centre Pavilion, plan drawing



This plan drawing details the overall spatial organization of the pavilion. It has been designed with an irregular polygonal footprint made up of alternating straight and inward-curving edges. As a space open to the public, the concave sections of the perimeter act as multiple points of entry, inviting visitors from multiple directions.



The elevation drawing gives more clarity to the shell character of the structure. The pavilion rises toward a central peak and descends towards the lower perimeter conditions. It becomes a shaded gathering space that provides shelter without being an enclosed room; this allows whoever is underneath it to remain connected to the surrounding landscape.

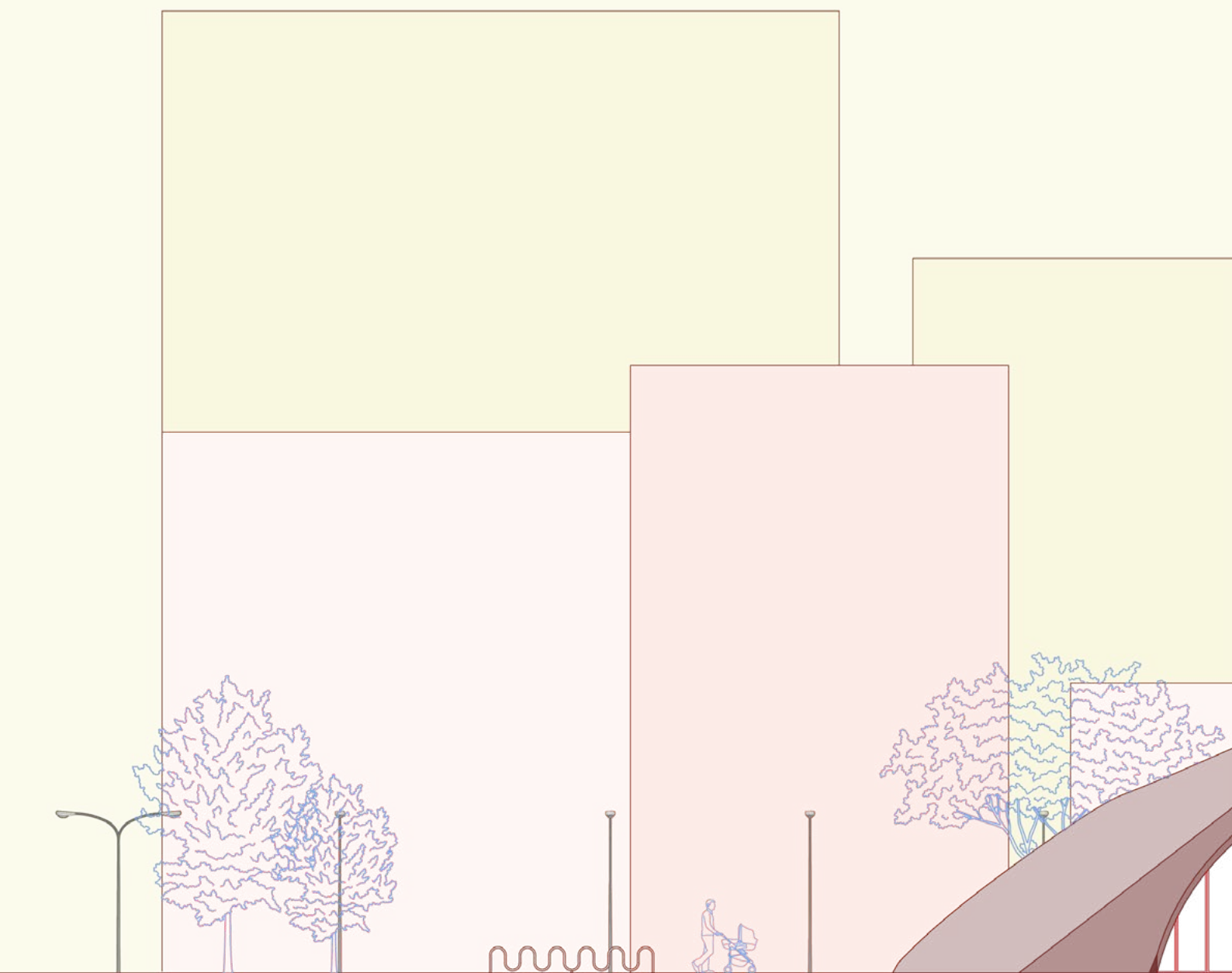
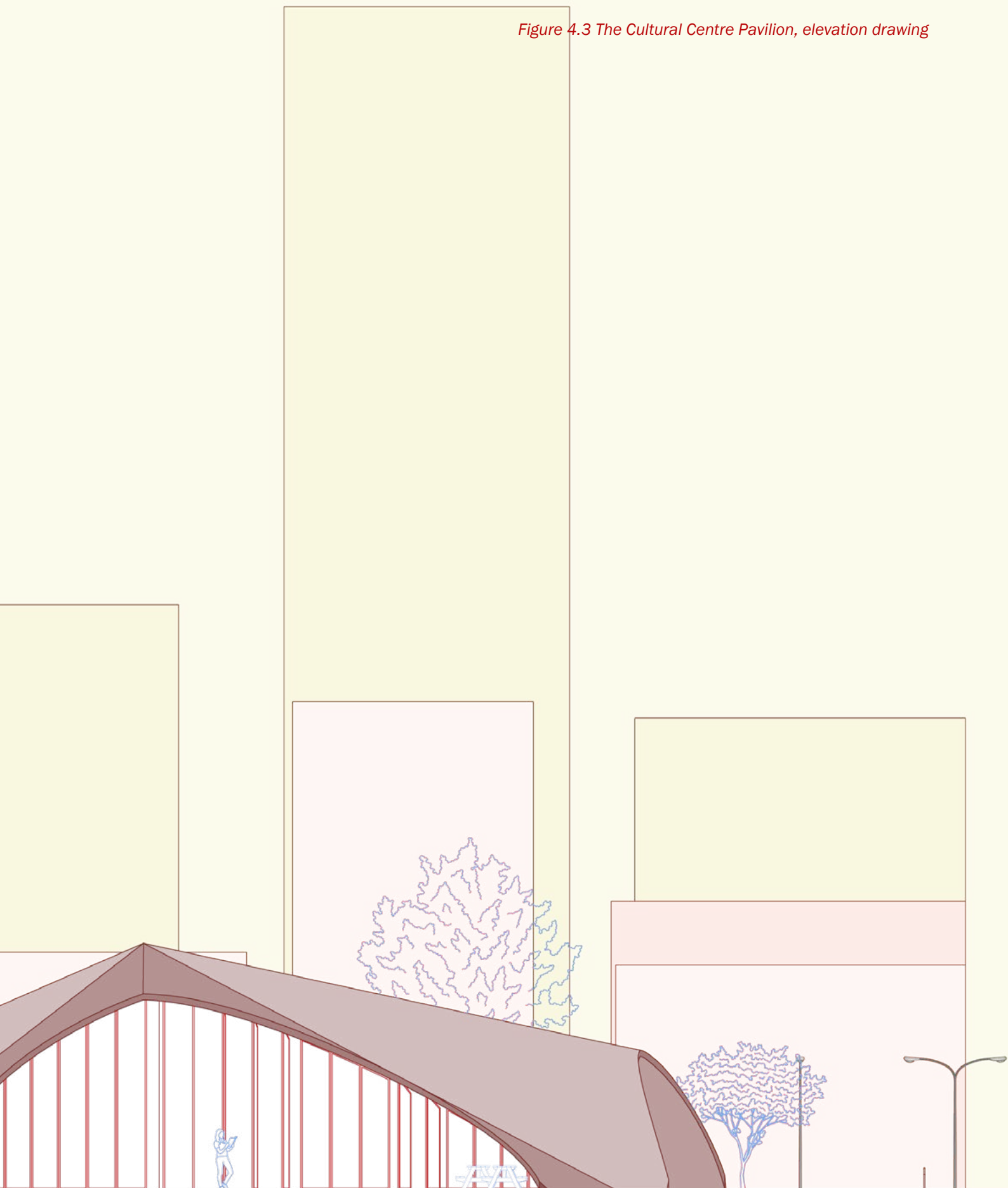


Figure 4.3 The Cultural Centre Pavilion, elevation drawing



The axonometric diagram illustrates how the overall shell is subdivided into sequences of individually fabricated panels that follow its changing geometry. The panels are organized into sequences that follow the geometry of the shell. This modular fabrication strategy emphasizes how the developed fabrication strategy produces manageable components such that the fabrication scale of the robotic setup does not have to correspond to the architectural scale of the design.

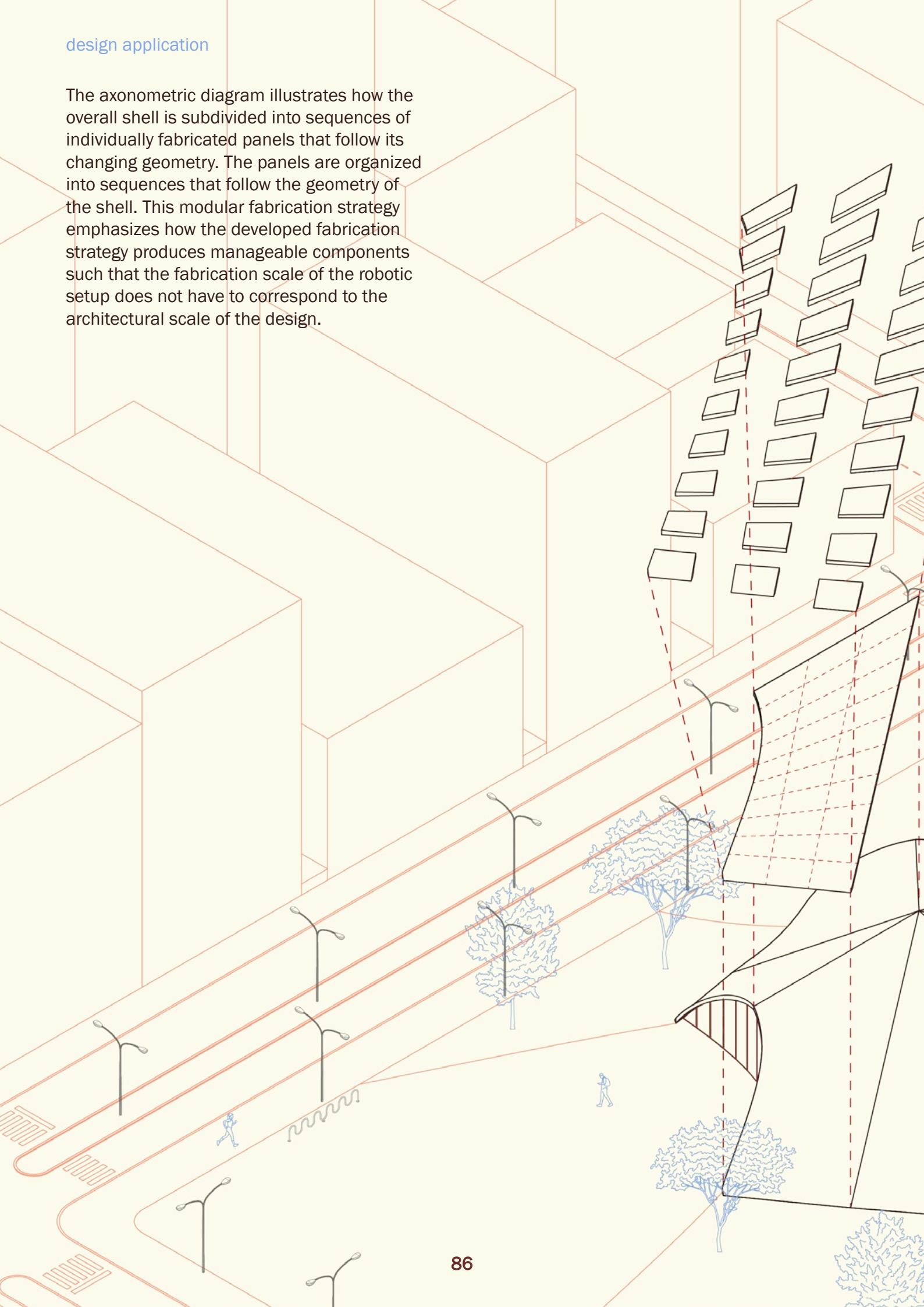


Figure 4.3 The Cultural Centre Pavilion, exploded axonometric drawing





Figure 4.5 The Cultural Centre Pavilion, exterior render





Figure 4.5 The Cultural Centre Pavilion, interior render



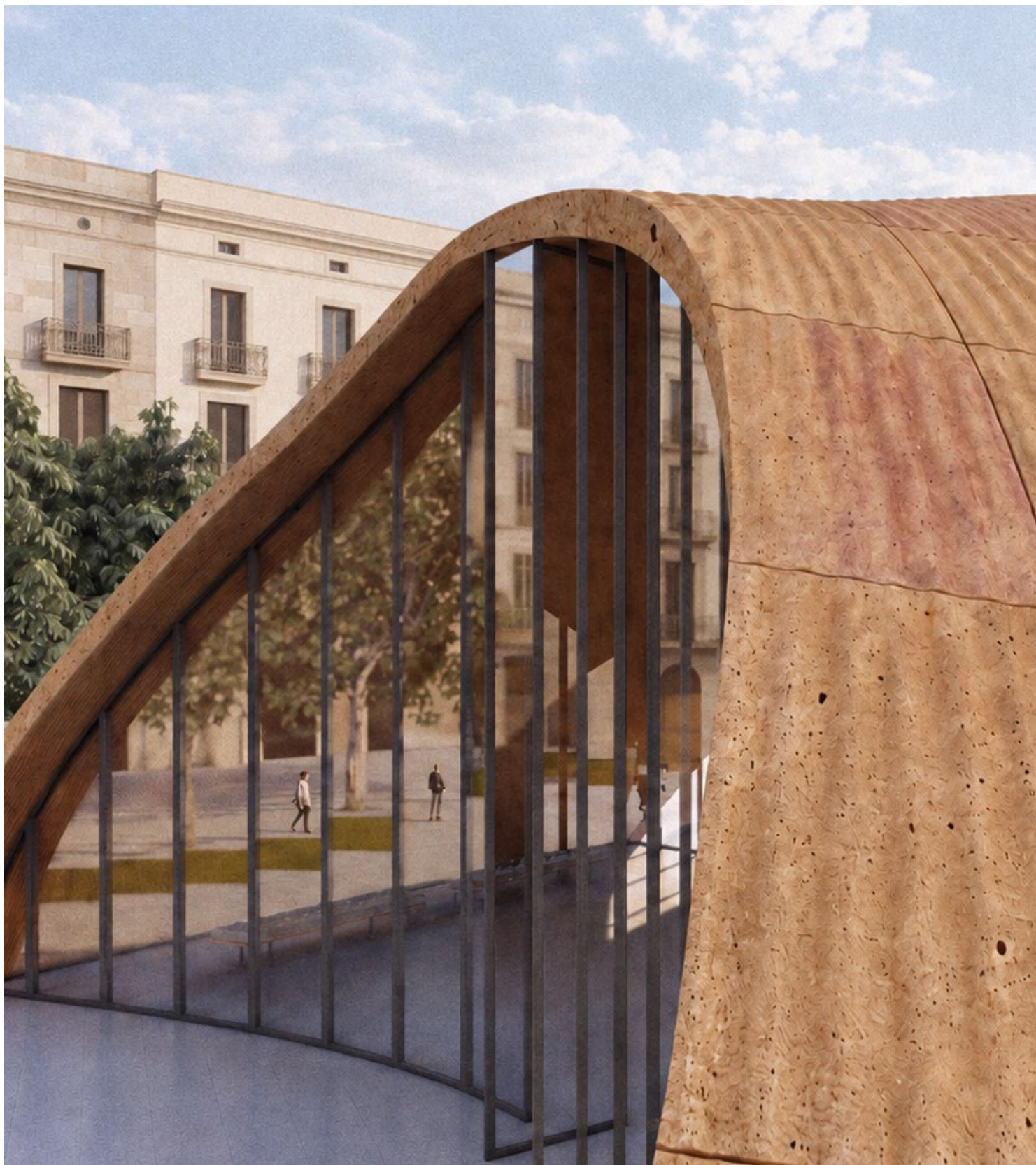


Figure 4.5 The Cultural Centre Pavilion, close-up view render



05

conclusion and future steps

As the investigation for this thesis comes to an end, it is important to reflect on all the learning outcomes and the outcomes and broader implications of the research. Even though the thesis covered a wide range of areas, from material to a new architectural language, the main contribution lies in the fabrication; namely, the dynamic casting system developed through the research. During the year spent investigating and designing this fabrication system, different questions came up that identified several opportunities for refinement and further investigation. This final chapter offers insights on these points of interest for future investigations that can push this research forward.



As the focal point of the thesis, there were many considerations about the fabrication process that emerged during its development. Some of these were practical. Being mindful of the time and resources available, there were steps of the fabrication system that were performed manually. Nevertheless, these manually performed operations identify opportunities for further automation. Given that the main goal of the fabrication was to achieve the dynamic casting process, the system setup used for the robot was simplified. Referring back to Chapter 3, in the developed setup, the end-effector (the attachment connected to the sixth axis of the robotic arm) also functioned as the upper frame of the formwork. This meant that for every component made, the end-effector had to be switched out. This can be time consuming, making this an area to be improved upon. With more time and computational development, a gripper end-effector could be used to grab onto the top frame, perform the movement and release it once secured to the bottom in order to swiftly move onto the next piece.

Another step that has potential to be automated is the actual casting of the geopolymer mix. Similarly to concrete pumps, a future iteration could investigate pumping the UEM geopolymer directly into the fabric formwork, reducing the time and labour associated with manual casting. This, however, is a tricky aspect to explore due to the variability in aggregate size. Looking at it in the simplest way, it might result in one of two directions: either explore the mechanics of the pump to allow more heterogeneity or create a filtering system that reduces the large aggregates in the UEM before turning it into a geopolymer. Lastly, and perhaps the most interesting part of the fabrication process, there is potential in the robotic movement itself. With the use of tools such as Grasshopper, the rotation axes and the robotic trajectory are aspects of the fabrication that can also be pushed further. Current investigations into non-planar 3D printing raise questions such as: **can the rotations of the robot be non-planar for a formwork like this? To what extent can the fabric be twisted/rotated?** These questions establish potential directions for future investigation.

From a material standpoint, this thesis investigated geopolymers at a preliminary level. As it was shown early on in Chapters 1 and 2, as a topic, geopolymers and the family of alkali-activated material is a whole area of research of its own. The decision to pursue AAMs as the material solution was instigated by the desire to repurpose waste. Although the geopolymerisation of UEM produced promising results, the material formulation and recipe development require further investigation. In fact, Chapter 2 explicitly mentions some of the



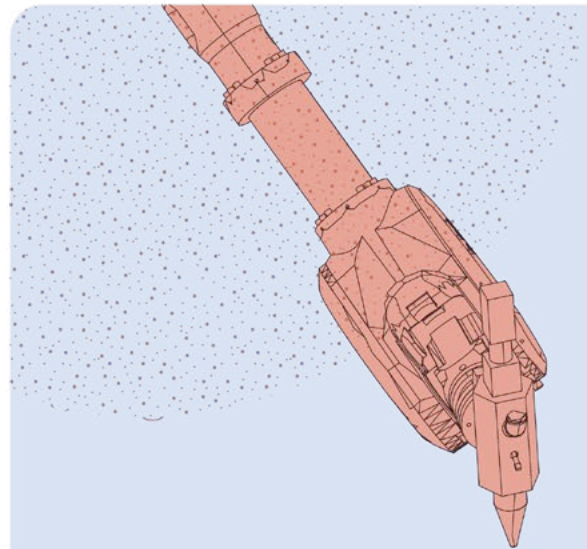


areas that are being investigated in regards to geopolymers.

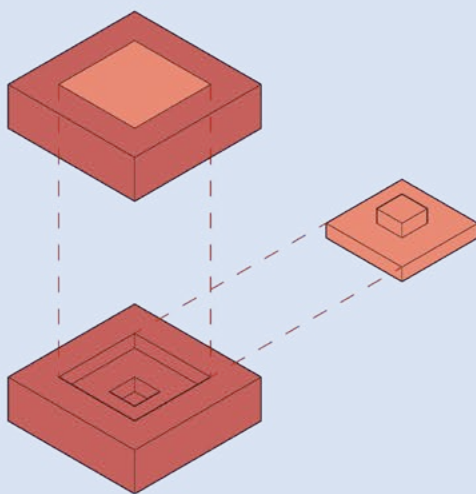
The research evolved organically, with the outcomes of each stage determining the subsequent course of investigation, which led to the exploration of more than one topic. On the one hand, this has left room for future steps to be made in this research, opening up new questions that could lead to new discoveries. On the other hand, it offered insights that impacted the collective knowledge of material and fabrication processes. This thesis has shown the potential in three main areas. The literature establishes the abundance of excavation and construction waste, while the experiments demonstrate its potential to be repurposed as a component of a new material system. It gives a second life to a material that would otherwise be discarded and relegated to low-value applications. Moreover, the research challenges fabrication practices dependent on highly homogeneous inputs by investigating a formwork-based process capable of accommodating greater material variability. This leads to the last point of contribution, which is a new architectural language, where both the material and formwork inform the design of the space. Rather than pushing a form onto a material with conventional moulds for casting, in the dynamic casting system, there is a dialogue between the formwork and the material that together design the piece, which is evident in both the geometry and the surface finish. As a system, it was designed considering both the qualities and limitations of the material, which creates a language that pushes the geometric possibilities.



excavation material is landfilled

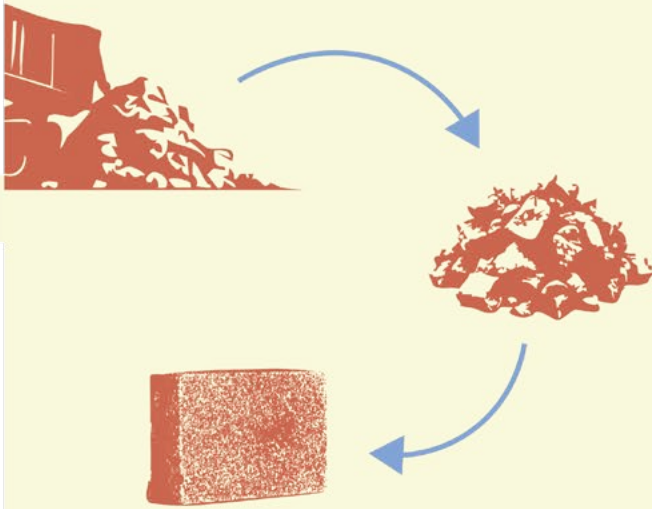


digital fabrication
with homogeneous material

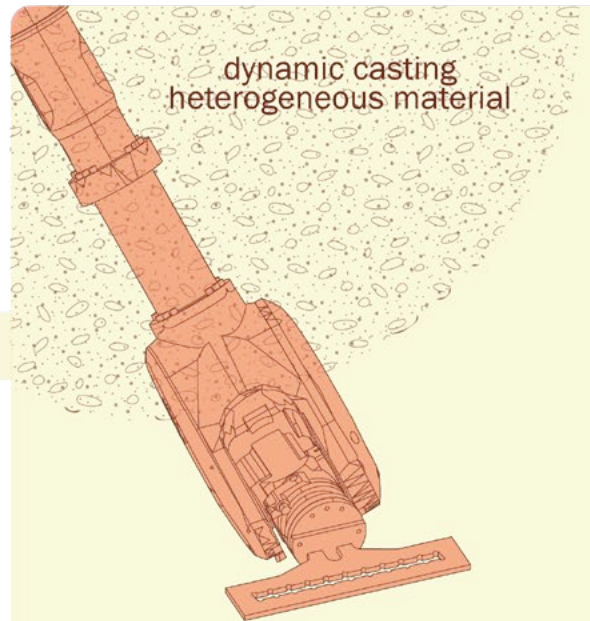


traditional casting system

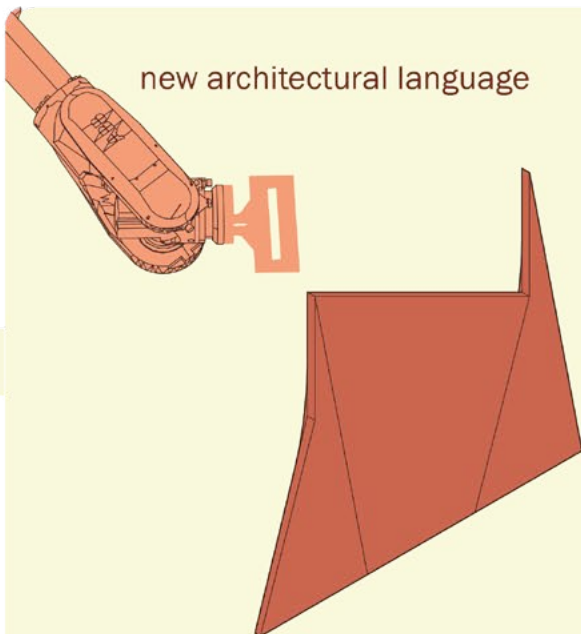
repurposing urban waste



dynamic casting
heterogeneous material



new architectural language



bibliography

Adesanya, E. et al. (2021) 'Opportunities to improve sustainability of alkali-activated materials: A review of side-stream based activators', *Journal of Cleaner Production*, 286, p. 125558. Available at: <https://doi.org/10.1016/j.jclepro.2020.125558>.

Baharami, H. et al. (2025) 'Fabricable Discretized Ruled Surfaces', *ACM Transactions on Graphics*, 44(3), p. 30:1-30:15. Available at: <https://doi.org/10.1145/3734519>.

Britannica Editors (2026) Polymerization | Definition, Classes, & Examples | Britannica. Available at: <https://www.britannica.com/science/polymerization> (Accessed: 29 July 2026).

Cao, T., Kotnik, T. and Schwartz, J. (2023) 'Smooth Poly-hypar Surface Structures: Freeform Shells Based on Combinations of Hyperbolic Paraboloids', *Nexus Network Journal*, 25(2), pp. 439–463. Available at: <https://doi.org/10.1007/s00004-022-00624-w>.

Chaudhury, M.K. (1996) 'Interfacial interaction between low-energy surfaces', *Materials Science and Engineering: R: Reports*, 16(3), pp. 97–159. Available at: [https://doi.org/10.1016/0927-796X\(95\)00185-9](https://doi.org/10.1016/0927-796X(95)00185-9).

Designing Buildings (2023) Alkali-activated materials (AAM). <https://www.designingbuildings.co.uk>. Available at: [https://www.designingbuildings.co.uk/wiki/Alkali-activated_materials_\(AAM\)](https://www.designingbuildings.co.uk/wiki/Alkali-activated_materials_(AAM)) (Accessed: 16 March 2026).

Dunant, C.F. et al. (2021) 'Good early stage design decisions can halve embodied CO2 and lower structural frames' cost', *Structures*, 33, pp. 343–354.

Elsayed, M. and Nehdi, M.L. (2017) 'Experimental and analytical study on grouted duct connections in precast concrete construction', *Materials and Structures*, 50(4), p. 198. Available at: <https://doi.org/10.1617/s11527-017-1056-0>.

European Commission (n.d.) Construction and demolition waste, European Commission. Available at: https://environment.ec.europa.eu/topics/waste-and-recycling/construction-and-demolition-waste_en (Accessed: 24 July 2026).

FRONT Materials (2021) '28 Incredible Statistics About Waste Generation', FRONT® Materials, 3 September. Available at: <https://www.front-materials.com/news/statistics-about-waste-generation/> (Accessed: 12 April 2026).

IPA (2019) IPA - Information Portal for Waste Assessment, Information Portal for Waste Assessment. Available at: https://www.abfallbewertung.org/repngen.php?report=ipa&char_id=1705_Boden&lang_id=en&avv=&synon=&kapitel=2>active=no (Accessed: 24 July 2026).

Joseph Davidovits (2020) Geopolymer Chemistry and Applications. 5th edn. Saint-Quentin: Geopolymer Institute. Available at: <https://www.geopolymer.org/learning/book-geopolymer-chemistry-and-applications/>.

Kim, K.W. et al. (2022) 'Fast-Curing Geopolymer Foams with an Enhanced Pore Homogeneity Derived by Hydrogen Peroxide and Sodium Dodecyl Sulfate Surfactant', *Minerals*, 12(7), p. 821. Available at: <https://doi.org/10.3390/min12070821>.

Komkova, A. and Habert, G. (2023) 'Environmental impact assessment of alkali-activated materials: Examining impacts of variability in constituent production processes and transportation', *Construction and Building Materials*, 363, p. 129032. Available at: <https://doi.org/10.1016/j.conbuildmat.2022.129032>.

Li, Z. et al. (2024) 'Design and construction of catenary-ruled surfaces', *Structures*, 59, p. 105755. Available at: <https://doi.org/10.1016/j.istruc.2023.105755>.

Malakatas, N.E. (2022) Eurocode 1: Actions on Structures – EN 1991-1-1. Ispra, Italy: Joint Research Centre, European Commission. Available at: https://eurocodes.jrc.ec.europa.eu/sites/default/files/2022-06/EN1991_1_Malakatas.pdf (Accessed: 2 August 2026).

Marín-Cortés, S. et al. (2024) 'Experimental characterization data on aggregates from construction and demolition wastes for the assistance in sorting and recycling practices', *Construction and Building Materials*, 435, p. 136798. Available at: <https://doi.org/10.1016/j.conbuildmat.2024.136798>.

Márquez-Muñoz, I.J. (2023) 'Casting concrete with fabric form-work: Active learning methodologies in architectural education Fabricando hormigón con encofrados textiles: Metodologías de aprendizaje activo en educación arquitectónica', *Advances in Building Education*, 7(1), pp. 43–50. Available at: <https://doi.org/10.20868/abe.2023.1.5087>.

Nurruddin, M.F. et al. (2018) 'Methods of curing geopolymer concrete: A review', *International Journal of Advanced and Applied Sciences*, 5(1), pp. 31–36. Available at: <https://doi.org/10.21833/ijaas.2018.01.005>.

Panesar, D.K. (2019) '3 - Supplementary cementing materials', in S. Mindess (ed.) *Developments in the Formulation and Reinforcement of Concrete* (Second Edition). Woodhead Publishing (Woodhead Publishing Series in Civil and Structural Engineering), pp. 55–85. Available at: <https://doi.org/10.1016/B978-0-08-102616-8.00003-4>.

Provis, J.L. (2018) 'Alkali-activated materials', *Cement and Concrete Research*, 114, pp. 40–48. Available at: <https://doi.org/10.1016/j.cemconres.2017.02.009>.

Provis, J.L. and van Deventer, J.S.J. (eds) (2009) *Geopolymers*. 1st edn. Woodhead Publishing. Available at: <https://shop.elsevier.com/books/geopolymers/provis/978-1-84569-449-4> (Accessed: 1 August 2026).

Qin, Y. et al. (2022) 'One-Part Alkali-Activated Materials: State of the Art and Perspectives', *Polymers*, 14(22), p. 5046. Available at: <https://doi.org/10.3390/polym14225046>.

Rehman, A.U. and Kim, J.-H. (2021) '3D Concrete Printing: A Systematic Review of Rheology, Mix Designs, Mechanical, Microstructural, and Durability Characteristics', *Materials*, 14(14), p. 3800. Available at: <https://doi.org/10.3390/ma14143800>.

Roosta, A., Zendehboudi, S. and Rezaei, N. (2025) 'Estimating advancing and receding contact angles for pure and mixed liquids on smooth solid surfaces using the PCP-SAFT equation of state',

Physical Chemistry Chemical Physics, 27(12), pp. 6031–6042.
Available at: <https://doi.org/10.1039/d4cp04054f>.

Scheder-Bieschin, L., Van Mele, T. and Block, P. (2024) 'Bending-active formwork systems for concrete shells – A classification and state-of-the-art review', Structures, 67, p. 106841. Available at: <https://doi.org/10.1016/j.istruc.2024.106841>.

Silva, R.V., de Brito, J. and Dhir, R.K. (2014) 'Properties and composition of recycled aggregates from construction and demolition waste suitable for concrete production', Construction and Building Materials, 65, pp. 201–217. Available at: <https://doi.org/10.1016/j.conbuildmat.2014.04.117>.

Sizirici, B. et al. (2021) 'A Review of Carbon Footprint Reduction in Construction Industry, from Design to Operation', Materials, 14(20), p. 6094. Available at: <https://doi.org/10.3390/ma14206094>.

Sönmez, N. and Kalfa, S.M. (2023) 'Investigation of Construction and Demolition Wastes in the European Union Member States According to their Directives', Contemporary Journal of Economics and Finance, 1(2), pp. 7–26.

Velimirović, L. et al. (2009) 'Ruled surfaces in architecture', Int. J. IT Secur., pp. 21–30.

Veres, C. and Tănase, M. (2025) 'Sustainable Shell Structures: A Bibliometric and Critical Review of Buckling Behavior and Material-Efficient Design Strategies', Applied Sciences, 15(17), p. 9394. Available at: <https://doi.org/10.3390/app15179394>.

Vogt, O. et al. (2019) 'Reactivity and Microstructure of Metakaolin Based Geopolymers: Effect of Fly Ash and Liquid/Solid Contents', Materials, 12(21), p. 3485. Available at: <https://doi.org/10.3390/ma12213485>.

Weisstein, E.W. (N/A) Hyperbolic Paraboloid. Wolfram Research, Inc. Available at: <https://mathworld.wolfram.com/HyperbolicParaboloid.html> (Accessed: 5 August 2026).

Yang, X., Loh, P. and Leggett, D. (2019) 'Robotic variable fabric formwork', *Journal of Computational Design and Engineering*, 6(3), pp. 404–413. Available at: <https://doi.org/10.1016/j.jcde.2018.10.001>.

Yankelevsky, D.Z. (2024) 'The uniaxial compressive strength of concrete: revisited', *Materials and Structures*, 57(6), p. 144. Available at: <https://doi.org/10.1617/s11527-024-02422-x>.

Zhang, Y. et al. (2024) 'Development of sustainable geopolymers with excavation soil powder as precursor: Cementitious properties and thermal-activated modification', *Journal of Building Engineering*, 91, p. 109745. Available at: <https://doi.org/10.1016/j.jobbe.2024.109745>.

